Privacy-Preserving Intelligent Intent-Based Network Slicing for IoT Systems

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***Abstract*—The proliferation of Internet of Things (IoT) services across diverse sectors such as healthcare, industrial IoT, and smart cities has introduced unprecedented complexity in network management and orchestration (MO). Contemporary MO systems are challenged to support the coexistence of IoT services with varying Quality of Service (QoS) requirements while ensuring end-to-end (E2E) performance across heterogeneous technologies and multiadministrator domain networks. To address these requirements, emerging technologies such as intelligent intent-based network slicing (I-IBNS) systems are increasingly employed, leveraging advancements in network automation, artificial intelligence (AI), and network slicing (NS). This article focuses on three critical challenges in developing I-IBNS systems for IoT: 1) E2E resource allocation; 2) privacy preservation in multiadministrator systems; and 3) E2E QoS assurance across heterogeneous networks and time-varying traffic. We propose a privacy-preserving I-IBNS framework, named Harmony Slice Master (H-SliceMaster), which integrates a knowledge management framework for privacy-aware data aggregation, an intent propagation mechanism for translating high-level intents into network configurations, and a novel promise and price network operation (PPNO) principle for optimizing E2E resource allocation while maintaining intradomain privacy. A proof-of-concept design of the H-SliceMaster is presented, utilizing deep Q-networks (DQNs) for intra-domain resource allocation and a centralized algorithm for optimizing E2E network slice deployment. Simulation results demonstrate that the H-SliceMaster efficiently satisfies various IoT applications and delay requirements. Moreover, the proposed system achieved an 20% and 50% reduction in system costs compared to a branching dueling Q-network (BDQ) and greedy algorithm, respectively.**

***Index Terms*—Deep reinforcement learning, intent-based networking (IBN), Internet of Things (IoT), management and orchestration (MO), network slicing (NS).**

I. Introduction

THE Internet of Things (IoT) has ushered in an era of unprecedented complexity in network management and orchestration (MO). This paradigm shift has introduced a multitude of services spanning diverse sectors, including industrial automation, healthcare, and smart cities, each presenting unique functional and nonfunctional requirements, e.g., in terms of Quality of Service (QoS) [1]. The open IoT ecosystem, leveraging virtualization and softwarization technologies enables the creation of E2E networks composed of heterogeneous domains, allowing for the fulfillment of service requirements. For instance, IoT services with intensive computational requirements can utilize virtual resources from cloud providers, hosting specific functions in the cloud environment. This flexibility extends to accommodating various design requirements, such as those posed by remote medical surgery services that demand stringent E2E delay and reliability constraints, even when spanning multiple countries. In such scenarios, the E2E network may comprise multiple administrative domains, each contributing to the fulfillment of strict QoS requirements [2]. The multiadministrator domains and heterogeneous technological approach introduce additional layers of complexity in terms of coordination, security, and performance management [3].

Recognizing these emerging complexities, the 3rd Generation Partnership Project (3GPP) introduced the concept of network slicing (NS) [4]. NS provides a robust MO framework, utilizing key functions such as the communication service management function (CSMF), network slice management function (NSMF), and network slice subnet management function (NSSMF). These functions are designed to integrate heterogeneous domains and facilitate coordination and collaboration between different network segments. For instance, an NS can contain several network slice subsystems (NSSs) including a radio access network (RAN), an edge network, and a cloud network. Each of these NSSs is managed by an NSSMF which coordinates with an NSMF to ensure E2E performance [5]. This holistic MO approach empowered by softwarization and virtualization technologies enables the coexistence of multiple E2E networks on the same physical infrastructure, each tailored to specific functional and nonfunctional requirements.

Moreover, network automation and intelligence have been widely adopted as a crucial part of future MO systems. Intent-based networking (IBN), the advent of network automation, offers a high-level, declarative approach to network management that enables network automation [6]. By emphasizing the desired outcomes rather than specific implementation details, IBN allows for autonomous network MO reducing service deployment times and minimizing configuration errors. Moreover, IBN defines specific functional components such as intent translation, resolution, activation, and assurance, enabling closed-loop service MO [7].

In these systems which we refer to by intelligent intent-based NS (I-IBNS), intents can be translated to an NS design that can be implemented over the IoT network [8]. These

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NS designs typically comprise service function chains (SFCs) consisting of multiple virtual network functions (VNFs). Moreover, intents are often associated with specific QoS requirements, which are derived through user interactions [9]. Realizing these intents within the IoT system necessitates NS deployment, which involves placing the SFC onto the physical network and allocating appropriate resources [10]. These resources may encompass computing capacities in edge and cloud domains, as well as radio resources in the access domain [11].

The development of I-IBNS systems introduce a complex balance between operational efficiency and privacy considerations in various scenarios. For example, in systems with multiple administrative domains, I-IBNS must handle intents with specific QoS requirements across diverse network components, such as access, transport, edge, and cloud infrastructure. This necessitates a MO framework with detailed knowledge of the deployed network, including resource utilization, capacity, and intent-driven QoS performance at each network element. artificial intelligence (AI)-powered resource optimization algorithms, which frequently form the core of these MO frameworks, typically rely on extensive amounts of potentially sensitive data [12]. This reliance on data introduces significant challenges in multidomain environments, where data sharing may be restricted due to privacy concerns [13]. Furthermore, I-IBNS may accommodate intents from multiple service providers, leading to a multitenant IoT network. In such environments, the MO framework may extend orchestration capabilities to each tenant, requiring the sharing of system data and granting varying levels of control over network operations, such as orchestration and resource management.

To address these challenges, privacy-preserving AI techniques such as federated learning (FL) and federated reinforcement learning (FRL) have emerged as viable solutions. These methods enable AI-driven optimization while maintaining data privacy by sharing AI models instead of raw data. For example, the work in [14] proposed a FRL approach for RAN networks, allowing collaboration between multiple vendors without exposing sensitive information. Although these algorithms offer data privacy benefits, they assume a level of trust between the participating entities, as the shared AI models operate with limited transparency and interpretability, making it difficult to predict their behavior and output. In multiadministrator environments, the exchange of AI models necessitates additional trust assurance mechanisms or explainability measures. Another stream of work focuses on the collaboration between MOs of different domains [15], [16]. For instance, Yousaf et al. [17] introduced the concept of Management and Orchestration as a Service (MANOaaS), which provides tenants with the ability to manage their network slices. However, this approach requires establishing strong policies to regulate data sharing and orchestration across different entities. Consequently, developing a privacy-preserving E2E MO framework becomes crucial to reconcile the tradeoff between system privacy and network optimization and to enable heterogeneous service models that include multiadministrator and multitenancy in future IoT MO systems.

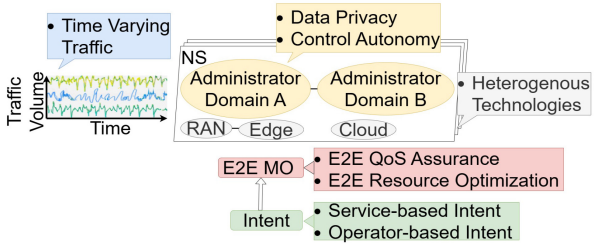


Fig. 1. Challenges and requirements of MO systems.

This article introduces the Harmony Slice Master (H-SliceMaster), a novel I-IBNS framework designed to harmonize performance optimization with stringent privacy safeguards while ensuring seamless coordination across diverse network domains to meet E2E intent requirements. We identify two distinct privacy dimensions: 1) data privacy and 2) control autonomy.

Data privacy pertains to the protection of system-related information, including resource availability and key performance indicators such as delay and throughput, which are critical for resource optimization but sensitive in multidomain and multitenancy environments. Control autonomy, on the other hand, refers to the privacy of system management, specifically the orchestration and resource allocation. This involves safeguarding access to system control mechanisms, and ensuring that tenants or service providers have appropriate but limited authority over resource orchestration without compromising overall system integrity.

Moreover, the H-SliceMaster distinguishes between two primary categories of intents: 1) service-based intents and 2) operator-based intents. Service-based intents originate from service providers and typically involve the instantiation of new services or modifications to existing ones. In contrast, operator-based intents are initiated by the IoT system owner or operator and aim to enhance overall system performance, such as optimizing resource utilization. Consequently, we identify the need for accommodating operator-based intents while preserving the privacy of the system domains.

The H-SliceMaster addresses these requirements as summarized by Fig. 1, by proposing efficient privacy-preserving mechanisms, such as the promise and price network operation (PPNO) principle, and the soft network control (SNC) mechanism.

A. Contributions

The primary contributions of this article are summarized as follows.

1) We propose the H-SliceMaster as a pivotal MO framework characterized by its ability to align E2E system performance, exemplified by network slices, with the intent objectives. The architecture framework of the proposed H-SliceMaster is presented, which incorporates IBN components within the NS management layers to allow for E2E network automation.
2) We propose two integral frameworks to the H-SliceMaster: a) a knowledge management framework

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and b) an intent propagation framework. The knowledge management framework is purposefully designed to address the complex task of privacy-conscious aggregation of data within IoT systems. It periodically collects key information inputs (KIIs) from heterogeneous systems, which are subsequently processed, enabling intelligent and automated network operations. The intent propagation framework focuses on translating high-level user intents into actionable subobjectives, which can be disseminated across different management layers. This process ensures that network orchestration policies are applied to system domains, thereby enabling seamless E2E intent fulfillment.

3) The PPNO principle is proposed to address the need for a delicate balance between system-wide optimization and data privacy. This balance is achieved by enabling system domains to share specific promises and associated prices with a centralized MO entity, diverging from traditional MO systems that reveal system domain orchestration capabilities, configurations, and performance metrics. This enables centralized MO entity to optimize E2E promises, where in each promise is translated into a policy within system domains. The PPNO framework ensures that each domain retains its privacy and autonomy while still achieving comprehensive E2E optimization objectives across the entire system.
4) The SNC mechanism is proposed to accommodate operator-based intents while maintaining system control autonomy. Rather than exerting direct control over system domains, the SNC mechanism achieves the desired system behavior by adjusting pricing strategies to reflect the operator's intents. This allows the over all system to adapt and align with operator objectives without compromising the autonomy of individual network domains.
5) We introduce four variations of the H-SliceMaster framework, strategically designed to explore the critical tradeoffs between privacy preservation and MO overhead. We analyze the bandwidth and latency requirements across different architectural configurations, providing a comprehensive framework for network operators to evaluate and design privacy-preserving MO strategies.
6) To validate the effectiveness of the proposed H-SliceMaster, we present proof of concept (PoC) algorithms. Specifically, we develop resource allocation policy leveraging Double Dueling deep Q-networks (D3QNs) for each system domain, which we refer to as intradomain policies. In addition, a centralized dynamic algorithm is proposed to optimize E2E NS deployment. The proposed dynamic NS deployment algorithm considers the time-varying characteristics of IoT traffic as well as time-varying resource availability. We conduct a detailed performance evaluation that includes several IoT applications such as Smart City, Smart Factory, and eHealth, with different E2E delay requirements. Moreover, we compare the H-SliceMaster against branching dueling Q-network (BDQ) and greedy algorithms.

This article is organized into eight sections. A comparison to related work is presented in Section II. The proposed H-SliceMaster framework is introduced in Section III. The H-SliceMaster frameworks and their evaluation are presented in Section IV. The PoC algorithms are discussed in Section V. Section VI carries the simulation testing and results. The discussion and future work is in Section VII. Lastly, the conclusion is in Section VIII.

II. Related Work

In this section, a state-of-the-art literature review on MO frameworks and E2E network optimization algorithms is provided with a focus on IoT-specific challenges and enabling technologies. As summarized in Table I, we compare recent frameworks proposed in the literature concerning eight aspects including the technology-enabling components such as NS, AI, and IBN. Moreover, the table includes the multiadmin and end-to-end (E2E) components, that indicate whether the work supports multiadministrator systems and E2E network segments. In addition, the table indicates the primary objectives, the domain, and whether each framework incorporates a specific optimization algorithm as part of its proposed methodology.

In the following, we discuss principal research gaps in the existing body of literature.

1) *Bridging the Gap Between Architectural Frameworks and Optimization Algorithms:* Research within the domain of management systems can be broadly categorized into two main streams: a) the development of management frameworks and b) the design of optimization-based systems. The first category primarily focuses on proposing management frameworks that address the various management layers and interactions within management entities. These frameworks are often tailored to specific technological domains, such as the 6G ecosystem [19], fog computing systems [39], and 5G multitenant networks [20]. In some instances, these frameworks are supported by PoC implementations [20] or simulations [38] to validate their operational effectiveness such as service creation time. However, many of these proposals fail to clearly define orchestration objectives, such as resource allocation optimization, leaving critical challenges, such as data privacy concerns unaddressed. For instance, while the study in [37] proposes a MO framework and discusses the deployment of SFCs, it falls short in illustrating practical scenarios where VNFs are distributed across different domains (e.g., edge and cloud), thereby overlooking the complexities involved in data sharing strategies. In contrast, the second research theme focuses on optimization algorithms, such as those presented in [18] and [11], which aim to achieve specific optimization objectives, often assuming the existence of an efficient MO framework. For example, the resource optimization algorithm proposed in [11] determines E2E configurations, including the transmission power of base

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TABLE I
OVERVIEW OF MO FRAMEWORKS: OBJECTIVES, DOMAINS, ALGORITHMS, AND ENABLING TECHNOLOGIES

Ref	Objective	Domain	Algorithm	NS	AI	IBN	Multi-Admin	End-to-End	Year
[18]	Intelligent orchestrator	RAN, Core	HA & RL	✓	✓	×	×	✓	2023
[11]	AI algorithm	RAN, Core, Backhaul	RL	✓	✓	×	×	✓	2023
[19]	E2E Framework	RAN, Core, Transport	None	✓	✓	×	×	✓	2023
[20]	NS Framework	RAN, Core, Transport	None	✓	×	×	×	✓	2023
[21]	NS scheme	RAN and Core	Stoch.	✓	×	×	×	✓	2023
[22]	Embedding algorithm	RAN, Edge, Core	HA & ILP	✓	×	×	×	✓	2021
[23]	E2E Architecture	Access, Edge, Transport, Core	RL	✓	✓	×	×	✓	2023
[24]	E2E Architecture	RAN, Core, Transport	None	✓	×	×	×	✓	2021
[25]	NS embedding problem	Access, Transport, Core	HA	✓	×	×	×	✓	2023
[26]	E2E NS Algorithm	Radio, Transport, Cloud	CQP	✓	×	×	×	✓	2022
[10]	NS algorithm	RAN, Edge, Core	RL	✓	✓	×	×	✓	2022
[27]	NS algorithm	RAN, Transport, Core	HA	✓	×	×	×	✓	2022
[28]	NS algorithm	Cloud	MBLP	✓	×	×	×	×	2021
[29]	E2E NS project	RAN	None	✓	✓	×	×	✓	2021
[30]	Deployment framework	RAN, Edge, Transport, Core, Cloud	None	✓	×	×	×	✓	2022
[31]	NS scheme	RAN, Fog, MEC, Cloud	AI-assisted	✓	✓	×	×	✓	2023
[32]	AI architecture	RAN, Edge, Core, Cloud	None	×	✓	×	×	✓	2022
[33]	AI NS architecture	Access	None	✓	✓	×	×	×	2022
[34]	AI NS framework	RAN, Cloud	FL	✓	✓	×	×	✓	2022
[35]	AI NS architecture	RAN, Edge, Core, Cloud	None	✓	✓	×	×	✓	2022
[8]	IBN architecture	RAN, Transport, Core	None	✓	✓	✓	×	✓	2022
[36]	E2E IBN framework	Access, Edge, Core	None	×	✓	✓	×	✓	2021
[37]	IBN architecture	Cloud-native domain	None	×	✓	✓	×	×	2023
[38]	IBN framework	Access, Core	None	✓	✓	✓	×	✓	2022
[39]	IBN architecture	Fog domain	None	×	×	✓	×	×	2022
[40]	IBN framework	RAN	DQN	✓	✓	✓	×	×	2022
[41]	Orchestration algorithm	NFV domain	algorithm	×	×	×	✓	×	2022
[42]	Orchestration framework	NFV domain	RL	×	✓	×	✓	×	2023

stations in the access domain and VNF placement in the core domain, which assumes full control and knowledge over system configurations. This observed pattern of the two principal themes in the literature is further corroborated by the data presented in the second and last columns of Table I, where literature works primarily oriented toward proposing frameworks tend to exclude algorithms from their methodological approach, while literature works that involve algorithms primarily aim at devising optimization algorithms without direct consideration of the intricacies associated with MO frameworks. Unlike previous works, our approach not only emphasizes optimization efficiency by benchmarking against advanced RL algorithms but also integrates a robust framework that provides essential data for optimization algorithms while simultaneously addressing privacy concerns.

2) *Overlooked E2E Orchestration Challenges:* Several recent works propose centralized algorithms for E2E resource optimization [10], [25], [37]. However, these approaches exhibit notable limitations in addressing critical orchestration requirements. A key oversight is the failure to account for the temporal disparities between different network domains. For example, the reconfiguration of RAN can occur with exceptional speed, often within millisecond intervals [43]. In contrast, the resource scaling or migration of VNFs in edge or cloud environments requires significantly more time, typically in the range of minutes [44]. Acknowledging these temporal discrepancies is crucial, especially in centralized E2E optimization algorithms that attempt to determine optimal configurations for the entire network in a single output. If service metrics such as latency or reliability deviate from their targets, these algorithms are ill-equipped to account for the varied operational timescales across different network domains. Therefore, introduces vulnerabilities in adaptability and responsiveness, particularly when rapid adjustments are needed to maintain optimal service performance. Furthermore, if intradomain changes are required such as software updates or resource expansion within system domains, these changes need to be communicated with the centralized MO entity. The architecture is highly sensitive to intradomain changes, requiring frequent coordination between the centralized MO entity and individual network orchestrators. Unlike previous proposals, the H-SliceMaster introduces a novel approach that preserves intradomain autonomy through the PPNO principle wherein domains maintain independent operation and commit only to specific promises, e.g., performance guarantees. This decoupling reduces coordination overhead with the centralized MO entity and allows domains to accommodate their unique operational characteristics, such as reconfiguration delays, without impacting the broader system.

3) *Importance of Considering the Time-Varying Traffic and Resource Availability:* Contemporary IoT environments are characterized by their inherent dynamism, necessitating support for diverse services with distinct stochastic patterns. While extensive research has been conducted on NS deployment and SFC embedding problems, a critical limitation persists: the majority of these studies treat traffic patterns and resource availability as static parameters [10], [25], [37]. This simplification can lead to suboptimal solutions that may fail to maintain QoS requirements throughout the NS lifecycle. Recent

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advancements, such as [45], have begun to acknowledge the time-varying nature of traffic and resource demand. However, these approaches typically consider only short-term variations, often limited to one-hour observation windows. Such restricted temporal analysis may not capture important traffic patterns that manifest over longer periods, such as daily variations. Our study addresses these limitations by incorporating long-term temporal dynamics in both traffic patterns and resource availability into the NS deployment optimization framework. We demonstrate the significance of this approach through comparative analysis with existing algorithms across multiple performance metrics. Furthermore, we evaluate our framework using diverse IoT services, such as eHealth, Smart City, and Smart Factory each exhibiting unique stochastic profiles derived from real-world traffic patterns [46].

4) *Exploiting Full Potential of Emerging Technologies:* It has become apparent that forthcoming MO systems will highly integrate emerging technologies such as NS and IBN, and AI/ML techniques. However, the integration of these advancing technologies into a unified framework has not yet been achieved. This observation is further underscored by the findings presented in Table I, which demonstrates that none of the examined studies has comprehensively addressed the integration of NS, IBN, and AI within an E2E context. In this work, we aim to fill this gap by proposing the H-SliceMaster framework, which integrates the NS management functions (MFs) and the IBN components. This is achieved by defining specific management layers and illustrating the purpose and functionality of the overall framework.

Table II provides a comparison between the proposed H-SliceMaster and traditional MO systems across multiple operational criteria. This comparison synthesizes the aforementioned discussion and highlights the distinctive features of each approach.

III. Proposed Privacy-Preserving Intelligent Intent-Based Network Slicing Framework

This section introduces the proposed privacy-preserving I-IBNS framework, namely, the H-SliceMaster. After that, we delve into the main components of the proposed H-SliceMaster, including, the knowledge management framework, the intent propagation framework, the PPNO principle, and the SNC mechanism.

A. Design Objectives and System Model

The proposed H-SliceMaster is an MO framework characterized by its ability to align E2E system configurations, exemplified by network slice instances, with the intent objectives. The proposed framework embodies the essence of autonomously orchestrating network resources according to the desired outcomes.

The utilization of NS technology in the proposed H-SliceMaster framework emerges as a strategic choice. NS offers a hierarchical and service-oriented management framework with the capacity to address the challenges inherent in heterogeneous collaborations among domains within the evolving landscape of IoT systems. Moreover, NS demonstrates a remarkable synergy with the fundamental objectives of IBN. It not only aligns with intent-driven configurations but also plays a pivotal role in mitigating the budgetary constraints associated with accommodating diverse functional and QoS requirements. This cost-effectiveness is achieved through the incorporation of softwarization and virtualization technologies.

The proposed H-SliceMaster comprises a network structure consisting of multiple NSs. In this context, all system resources, including communication links and computing servers, are affiliated with one or more NSs. These NSs serve as the foundation for connecting virtual and/or physical resources, thereby forming logical networks.

Within the H-SliceMaster framework, two distinct categories of NSs exist: 1) those affiliated with specific services and 2) those designated as backup resources, denoted as NS_X. The inclusion of NS_X follows a standard network design principle aimed at enhancing system reliability and facilitating the provision of QoS assurances [47]. Additionally, NS_X can assume active or standby configurations, with the H-SliceMaster accommodating the presence of one or multiple NS_X slices. Furthermore, NS_X can be presented as a NSS, allowing for its association with other NSSs to form a comprehensive NS.

The system model encompasses an IoT network characterized by a hierarchical administrative structure and heterogeneous implementation architecture. At its core, a primary network operator maintains ownership and operational control over the majority of the network infrastructure. However, certain network segments, e.g., a cloud computing domain, may fall under the administrative purview of external system administrators. The network operator facilitates intent-based service provisioning, enabling the dynamic instantiation of network slices, each tailored to deliver specific services with distinct requirements. Hence, the network operator owns and deploys the H-SliceMaster for the MO of the system resources and for extending MO capabilities to tenants, e.g., service owners. Moreover, all intradomains participate in the PPNO network operation mechanism. This architectural framework establishes a multitenant environment where diverse services coexist within the heterogeneous network infrastructure.

The design objective of the proposed H-SliceMaster framework is to achieve a delicate balance between E2E resource management optimization and the privacy of each intra domain. To contend with the intricate balance between privacy preservation and system performance, the framework introduces the PPNO principle. This principle is implemented as follows: each system domain exposes a promise and price list (PPL), encompassing details about supported QoS metrics, network functionalities, and their corresponding costs. In contrast to traditional frameworks that entail the disclosure of network configurations to the MO system, the PPNO principle permits systems, such as RAN systems or distributed edge networks, to orchestrate their respective domains while sharing PPLs that enumerate supported promises (e.g., QoS

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TABLE II
COMPARING THE HARMONY SLICE MASTER TO THE MO FRAMEWORKS IN THE LITERATURE

Criteria	Harmony Slice Master	Traditional MO
Optimization objective	End-to-end *policy chain* optimization	End-to-end *configuration settings/node selection* optimization
Data sharing	*Privacy-preserving* of system domains	Network capability and configuration *exposure*, and system *data sharing*
System domain autonomy	Allows for *independent* System domain autonomy and evolution	Very *sensitive* to domain changes
Multi-administrator Domains and tenant MO systems	*Privacy-preserving* management domains and control autonomy	Requires *strict security protocols and enforcement* for data sharing and system control, requires continuous coordination between the centralized MO and domain-specific and tenant-specific MO systems.

ranges) and their associated costs with an E2E MO entity. Consequently, the E2E MO entity optimizes E2E promise and price values (PPVs) where each PPV represents a selected Promise and associated Price from a PPL. This innovative approach facilitates the preservation of intradomain data privacy while steadfastly supporting the overarching goal of E2E optimization.

Furthermore, the H-SliceMaster framework facilitates the autonomous integration and enforcement of network operators' intents within multitenant environments while maintaining the confidentiality of domain-specific system configurations. Consider a scenario where network operators need to initiate traffic migration from specific edge servers. In traditional approaches, based on MO service agreements this process typically requires direct communication and negotiation with service owners, or requires network operator to configure a centralized MO entity possessing comprehensive control and visibility over the underlying network domains. Both approaches present significant challenges—the former introducing operational delays through prolonged negotiations, and the latter requiring configuration tasks and exposure of intradomain system configurations.

The H-SliceMaster addresses these limitations through the introduction of the SNC mechanism. This novel approach translates operator intents into a dynamic framework of incentives, penalties, and resource-specific cost adjustments. By modulating these economic parameters, the system induces desired behavioral changes in resource utilization patterns autonomously, without direct intervention in domain-specific configurations. This methodology effectively preserves control autonomy and operational privacy of intradomain systems while eliminating the need for time-consuming negotiation processes. The SNC mechanism thus represents a paradigm shift from direct control to incentive-based orchestration, enabling efficient intent realization while maintaining system privacy and autonomy.

B. Framework Architecture

The framework of the proposed H-SliceMaster is depicted in Fig. 2. The framework comprises three distinct layers, specifically the management view layer (MVL), the network slice management layer (NSML), and the IBN layer (IBNL). The MVL encompasses four primary functions, namely the system view function (SVF), the NSS view function (NSSVF),

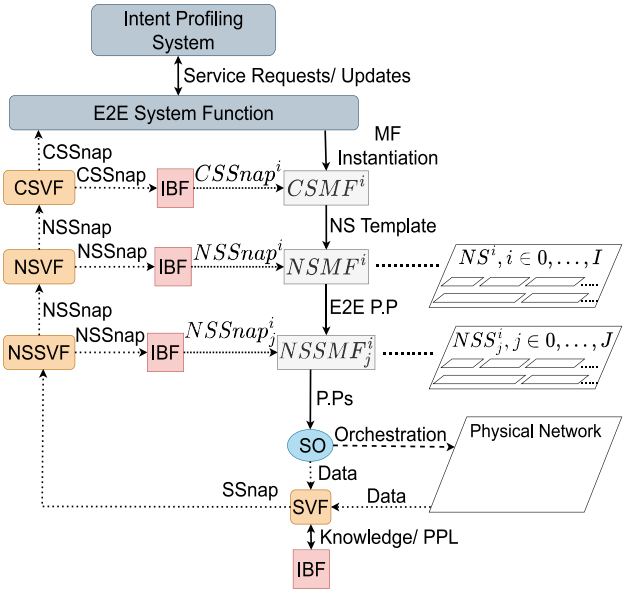


Fig. 2. Proposed harmony slice master framework.

the NS view function (NSVF), and the communication service view function (CSVF). Conversely, the IBNL is anchored by the intent-based networking function (IBF). In parallel, the NSML incorporates various NS MFs as defined by the 3GPP, specifically the CSMF, NSMF, and NSSMF [48].

Furthermore, the framework is augmented by an intent profiling system, serving as an intermediary interface connecting users with the IoT system. The intent profiling system translates the high-level user intent into a service request. Consequently, an E2E system function communicates the intent with any of the IBFs in the layered H-SliceMaster framework. For instance, an operator intent targeting a specific access domain in a geographical location associated with an NSS should be communicated with the IBF of the NSS layer. This requires the E2E system function to have direct interfaces communicating with the IBFs of the H-SliceMaster framework. In addition, an intent classification model needs to be deployed within the E2E system function to aid in directing intents to the correct system, e.g., the system responsible for MO of associated resources. This implementation ensures that the H-SliceMaster can effectively address intents without altering its original architectural framework. Additionally, the E2E system function is responsible for instantiating MFs.

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The MVL and IBNL layers oversee network data processing and system governance by conveying information such as orchestration costs to the NSML. The MVL handles basic data analytics, processing information from system domains and providing an E2E perspective. Within these system domains, IBFs develop tasks like training AI-based policies that support PPLs. At higher management levels, IBFs also serve as middleware between the MVL and NSML, enabling the admission of network operators' intents to be integrated into different management layers, such as the communication service (CS), network slice, and NSS layers. For instance, if the network operator anticipates increased service demand in certain geographical areas, an intent can be introduced within the IBF connected to the NSMF or the NSSMF to raise resource allocation costs in those locations, prompting the NSML to optimize resource allocation accordingly.

Notably, the functions within the MVL and IBNL remain agnostic to the number of instantiated network slices. Conversely, the number of instances for CSMF, NSMF, and NSSMF is contingent upon the quantity of instantiated services, NSs, and NSSs, respectively. As visually depicted in Fig. 2, the data flow within the H-SliceMaster traverses the MVL, while system orchestration is executed through the NSML and system orchestrators (SOs).

In the following, we discuss in detail the main enablers of the H-SliceMaster framework, namely, the PPNO, the knowledge management framework, the intent propagation framework, and the SNC mechanism.

C. Promise and Price Network Operation Principle

The objective of the PPNO principle is to enable E2E MO while accommodating the distinct operational characteristics of network domains, upholding intradomain MO autonomy, and preserving intradomain data privacy. PPNO operates through a well-defined mechanism within intradomain systems, consisting of three primary functions: the IBF, the SVF, and the SO.

Within the PPNO principle, the SVF processes raw system data and disseminates knowledge and insights to the intradomain IBF. Armed with this knowledge, the IBF curates a promise and price table (PPT), comprising commitments, e.g., promises, in the form of supported QoS metrics, network functions, and relevant details like geographical coverage. Notably, these commitments are closely associated with costs, e.g., prices, that reflect resource allocation and system operation expenses. Furthermore, the PPT encompasses resource management and allocation policies to uphold each PPV.

The actual design and optimization of E2E network slices occur at the NSML. The primary objective of the NSML functions is to optimize PPV selection across the entire network, and hence, the E2E PPVs. In this intricate process, the intradomain systems exhibit dynamic responsiveness to fluctuations in traffic and system conditions, as long as these configuration changes align with the policies linked to the selected PPV. Importantly, the NSML remains oblivious to the specific mechanisms employed by the system domains to uphold their promises. It cannot directly control individual configurations within intradomain systems. This functional isolation serves a dual purpose: it facilitates independent evolution for each system, preserving the privacy of system domains while accommodating multiadmin domain scenarios. This segregation of responsibilities within the PPNO framework offers a robust foundation for E2E MO while ensuring the integrity of intradomain operations.

Fig. 3 provides a visual representation of the end to end system model, comprising distinct domains, namely the device/sensor domain, RAN domain, transport domain, edge domain, and cloud domain. Various system domain orchestrators are deployed to manage the system resources by interacting with the H-SliceMaster. Moreover, the figure illustrates the functionality of the PPNO principle.

D. Knowledge Management Framework

The knowledge management framework plays a pivotal role in facilitating the functionality of the proposed H-SliceMaster by enabling the transfer of knowledge across the layered management framework. Within this framework, network functions are empowered to process data and simultaneously communicate data inference results to higher-level MFs. This approach, characterized by contained data processing mechanisms, effectively supports the privacy-preserving attributes of the proposed H-SliceMaster.

Specifically, the framework leverages the utilization of CS snap shots (CSSnap), network slice snap shots (NSSnaps), and system snap shots (SSnap) as data templates. These templates comprise KIIs derived from system domains. Following the collection of these snapshots, the MVL processes them and conveys the results to the IBL for further analysis. Subsequently, this knowledge is shared with the NSML for E2E resource optimization.

Within each system domain, the SVF processes raw system data, encompassing resource utilization and service performance metrics. This processed knowledge is then transmitted to the IBF to formulate the PPL accordingly. The SVF incorporates the PPL into an SSnap, which is then shared with the NSSVF. The NSSVF processes SSnaps from all systems within the IoT network, creating individual NSSnaps for each NSSI. These NSSnaps encapsulate the E2E PPL of the system domains included in each NSSI. The NSSnaps are subsequently forwarded to the IBF for further processing. Ultimately, they are conveyed to the NSVF, which assembles a per NSI NSSnap, accumulating the E2E PPL of the contained NSSs. Finally, a CSSnap is constructed for each CS instance (CSI), representing an abstracted E2E status of a CSI.

The knowledge management framework further enables real-time monitoring of network status by continually inquiring about these snapshots. This continuous monitoring process ultimately empowers E2E knowledgeable orchestration within the proposed H-SliceMaster.

E. Intent Propagation Framework

The intent propagation framework in the proposed H-SliceMaster enables the translations of intent within the NSML functions such that high-level user intent is optimally realized and sustained within the IoT system.

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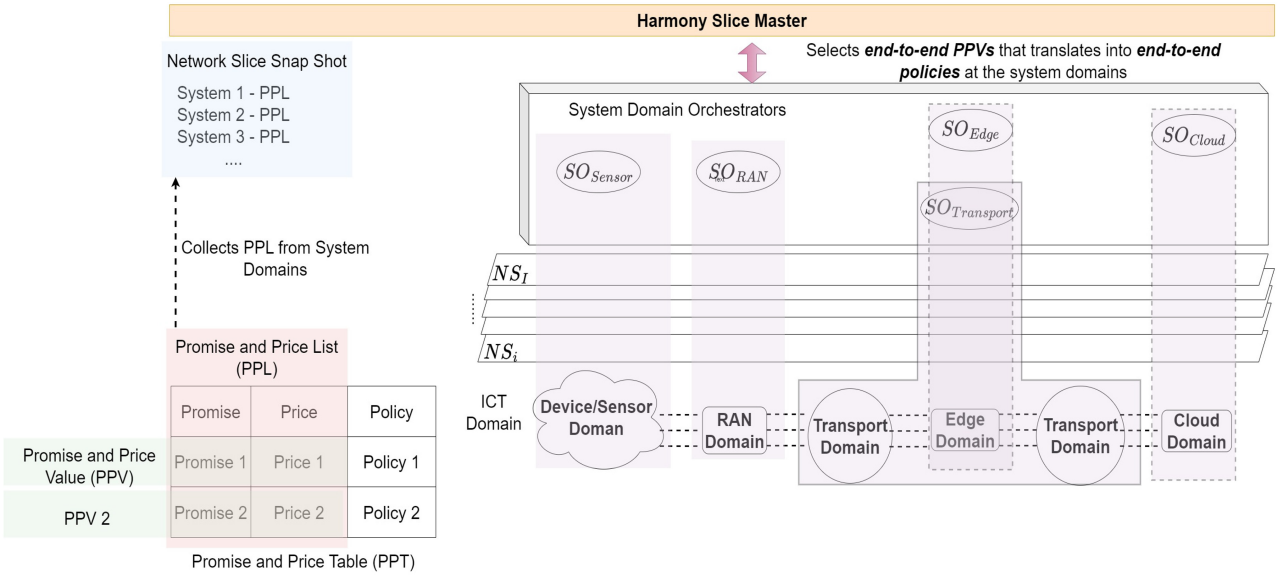


Fig. 3. PPNO principle.

Particularly, intents can be incorporated using the intent profiling system. The intent profiling system translates service-based intent into a CS request. Once a CS request is received, the E2E system function instantiates a CSMF which communicates with an IBF to design an NS template that satisfies the CS request. Consequently, the E2E system function instantiates an NSMF which processes the NS template and translates it into one or more NSS templates. In the design process, the NSMF will communicate with the IBF to enquire about available NSSs and their associated NSSnaps. Each of the NSS will be managed by a NSSMF. Consequently, the NSSMFs will communicate the orchestration commands, e.g., selected PPVs, with the SOs. Lastly, the SOs will deploy the policies associated with the selected PPVs.

The NSML ensures the service level of the instantiated NS such that it aligns with the original intent requirements. This is done by the continuous processing of knowledge snapshots and the dynamic optimization of E2E PPVs.

F. Soft Network Control Mechanism

When an operator-based intent is received, the SNC mechanism facilitates its processing through one or more IBFs. These IBFs serve as specialized translation entities, transforming operator-based intents into corresponding economic parameters within the PPV framework. Specifically, the IBFs modulate the pricing structure by either increasing or decreasing specific resource costs, e.g., prices, to reflect the operator's strategic objectives. This price modulation mechanism creates a dynamic feedback loop wherein the NSML, during its optimization of PPV selection, naturally aligns its resource allocation decisions with the operator's intent. This approach enables indirect yet effective control over system behavior while maintaining the autonomous nature of the management framework.

In the following section, we present a PoC study addressing an intent-oriented NS embedding problem. This problem, typically referred to as the VNF embedding problem [45], is examined with a particular emphasis on ensuring a QoS requirement, which serves as a key aspect of the intent.

IV. H-Slice Master Frameworks Evaluation

In this section, we discuss four implementations of the H-SliceMaster, domain-based, NSS-based, and NS-based the performance of the MO framework. These frameworks differ in their capabilities regarding privacy-preservation MO and complexity. Afterward, we present a use case study to evaluate each framework's scalability and implementation feasibility.

A. H-SliceMaster Frameworks

We introduce four distinct frameworks within the H-SliceMaster paradigm, each designed to provide specific privacy-preserving capabilities: the Domain-based, NS-based, NSS-based, and E2E H-SliceMaster frameworks. Fig. 4 illustrates these variations, highlighting their structural differences and integration with existing 3GPP MFs, including the CSMF, NSMF, and NSSMF. Note that the architecture of the E2E H-SliceMaster is shown in Fig. 2.

At the core of the H-SliceMaster architecture is the introduction of two essential layers: the MVL and the IBNL. Consequently, the Domain-based framework, the fundamental model of H-SliceMaster, introduces the functions of the MVL and IBNL within system domains, namely, the SVF and the IBF. This enables the PPNO principle to operate within system domains and the E2E MO entity. In this framework, each system domain shares PPLs with the E2E MO entity, which then formulates the E2E policy chain.

The NSS-based framework extends this architecture by deploying the MVL and IBNL within NSSs. Here, the PPLs associated with NSSs are shared with the NSSMF, which assumes the role of policy chain management. Additionally, the E2E MO entity can leverage NSS PPLs to optimize and integrate NS operations. Since the MVL and IBF layers

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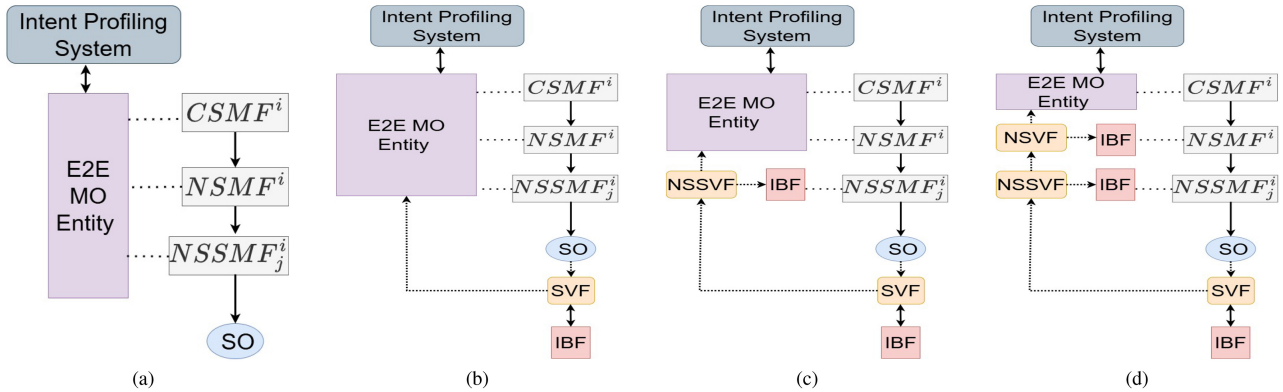


Fig. 4. MO frameworks. (a) Centralized MO, and H-Slice master. (b) Domain-based. (c) NSS-based. (d) NS-based.

facilitate the PPNO principle, the SNC mechanisms can also be effectively implemented within NSSs and system domains.

It is important to note that the accessibility of PPLs within the NSS-based framework is contingent on use case requirements and business agreements. The E2E MO entity may or may not have access to PPLs associated with specific system domains. For instance, in a multioperator network, where a business entity manages an NSS that integrates diverse technological domains—such as RANs and edge systems—an operator may choose to withhold system individual PPLs from the E2E MO entity. This ensures system privacy and limits the E2E MO entity's orchestration scope to NSS-level PPLs. Consequently, the E2E MO entity would be unable to autonomously integrate a particular system domain into another NS or dictate its specific performance parameters.

The NS-based framework further expands this model by integrating the MVL and IBNL at the NS level, introducing the NSVF alongside the IBF.

For comparison, Fig. 4(a) represents traditional MO architectures as proposed in the literature. These typically employ a centralized MO entity interfacing with 3GPP MO functions. Several works, such as [20] and [26], propose frameworks featuring an E2E Slice Manager, which integrates topology engines, metric collectors, and other orchestration mechanisms. Similarly, Liu et al. [23] introduced an E2E orchestrator that interacts with domain managers, while [34] defines an interdomain MO (IDMO) framework comprising a centralized component with full-scale slice MO capabilities. These models predominantly rely on a centralized MO entity, equipped with comprehensive system knowledge and orchestration control. However, their differences primarily lie in module implementations and E2E algorithmic variations.

In the following section, we provide a comparative analysis of the privacy features of the H-SliceMaster frameworks and those of the centralized MO frameworks commonly found in the literature.

B. Frameworks Evaluation Study

In this section, we evaluate the H-SliceMaster and centralized MO frameworks in two categories. First is the privacy features of the framework which include data privacy and control autonomy. The second is the implementation feasibility of the framework including the scalability study.

1) Privacy Features Study: Table III presents a comparative analysis of the centralized MO framework and the four H-SliceMaster frameworks—Domain-based, NSS-based, NS-based, and E2E H-SliceMaster—focusing on their privacy features.

System Exposure: A key aspect of this comparison is system exposure, which refers to the extent of data that underlying MFs and system domains must share with the E2E MO entity. In the centralized MO framework, data sharing is extensive. A data pool or data fabric maintains a repository of network-related information, including performance indicators, system capabilities, and other operational metrics. The E2E MO entity has direct access to this data, integrating it into MO optimization algorithms to enable resource allocation across the network.

Conversely, the H-SliceMaster frameworks introduce varying degrees of data exposure to enhance privacy. The Domain-based framework limits exposure to CSs, NSSs, and NSs, excluding system domain-specific data. This is possible by employing the SVF and IBF, which enable resource optimization through the PPNO principle. The NSS-based framework employs the MVL and IBNL at both the system domain and NSS levels. As a result, only NS and CS data are shared with the E2E MO entity. The NS-based framework further restricts data exposure, sharing only CS data with the E2E MO entity, as the PPNO principle is applied at the system domain, NSS, and NS levels. CS data may include service fee structures, fee adjustments, user priority levels, access control policies, and QoS metrics such as latency, security, and bandwidth. The E2E H-SliceMaster framework adopts a fully decentralized MO model, eliminating the E2E MO entity. Instead, MO tasks are distributed among specific MFs, and data sharing is governed by a knowledge management framework, ensuring maximal privacy.

Control Autonomy and SNC Capability: The degree of control autonomy and the applicability of SNC within a framework is directly tied to its MO principles. In the centralized MO framework, the E2E MO entity exercises direct system configuration and resource allocation, eliminating system-level autonomy. Consequently, SNC is not applicable in this model. In contrast, the H-SliceMaster frameworks allow for SNC

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TABLE III
COMPARISON OF MO FRAMEWORKS

Framework	System Exposure	Control Autonomy & SNC Capability	Privacy & Control Autonomy Level	Use Case Example
Centralized	All system data	Not applicable	Low	Simple networks with no privacy concerns
Domain-based	NSSs, NSs, and CSs data	Applicable to domains	Private domains	Multi-vendor
NSS-based	NSs and CSs data	Applicable to NSSs and domains	Private NSSs and domains	Multi-operator
NS-based	CSs data	Applicable to NSs, NSSs, and domains	Private NSs, NSSs, and domains	Multi-service
E2E H-SliceMaster	No data exposure	Applicable to all system layers	High	Multi-tenancy, multi-administrator, multi-vendor

implementation at the layers where MVL and IBNL are introduced, as specified in Table III.

Privacy and Control Autonomy Level: Privacy and control autonomy levels are directly related to the extent of data exposure and a system's ability to make autonomous resource management decisions. The centralized MO framework exhibits low privacy and control autonomy due to its reliance on centralized data access and orchestration. The E2E H-SliceMaster framework, in contrast, provides high privacy and control autonomy, as orchestration is fully decentralized, and data exposure is minimized. The Domain-based, NSS-based, and NS-based frameworks offer intermediate levels of privacy and autonomy, depending on the deployment of MVL and IBNL.

Use Case Example: Additionally, Table III also provides representative use case examples for each framework.

The centralized MO framework is well-suited for simple IoT networks where privacy is not a primary concern. For example, in a scenario where the entire IoT system is owned by a single business entity, all system data can be aggregated into a centralized data pool. Security measures such as access control and intrusion detection systems (IDSs) may be sufficient to maintain security.

The Domain-based framework is applicable in multivendor environments, where privacy between collaborating vendors is a concern [14]. The NSS-based framework is ideal for IoT systems involving multiple network operators, where an operator manages an NSS spanning multiple system domains (e.g., RAN and edge systems). This framework enables privacy-preserving coordination and E2E resource optimization by allowing operators to share only PPLs without exposing detailed system data. The NS-based framework is suitable for fully service-oriented NSs, where privacy preservation at the NS level is essential.

The E2E H-SliceMaster framework is the most comprehensive in addressing privacy concerns across an entire network, making it ideal for multitenancy systems. In such environments, an IoT system may serve multiple business entities (e.g., vertical industries) that require enhanced privacy and control autonomy. For example, by implementing the E2E H-SliceMaster, CS data—such as user access control policies and monetary fee adjustment mechanisms—is associated with abstracted NSS snapshots shared via the NSVF, ensuring privacy-preserving operations.

2) Implementation Study: The H-SliceMaster introduces a sophisticated framework to specify MFs' responsibilities and optimize MO workflow within a fully automated, privacy-preserving system. This objective necessitates the introduction of multiple MFs, resulting in signaling overhead generated by interfunction communications. This continuous message transmission signaling mechanism facilitates critical resource allocation decisions through the exchange of NSSnaps and PPLs. Consequently, the framework's complexity exhibits a direct proportional relationship with the number of MFs, where an increase in MFs amplifies the signaling overhead. A larger number of MFs can affect the latency of signaling messages received by the 3GPP MFs, causing a delay in resource MO decisions. Furthermore, the volume of network slices influences the magnitude of exchanged signaling messages.

TABLE IV
SIMULATION SETTINGS FOR THE H-SLICEMASTER EVALUATION

Parameter	Value
Number of MFs per Framework	
Domain-based framework	1 IBF, 1 VF
NS-based framework	2 IBFs, 2 VFs
NSS-based framework	3 IBFs, 3 VFs
E2E H-SliceMaster framework	4 IBFs, 4 VFs
Latency per MF	$\mathcal{N}$(1 µs, 0.5 µs) [49]
Bandwidth per message	$\mathcal{N}$(13 KB, 2 KB)
Message exchange rate per MF per hour	$\mathcal{N}$(6, 2)

In this section, we analyze the complexity of the H-SliceMaster frameworks associated with the MO signaling overhead, including latency and bandwidth requirements. Additionally, we evaluate the scalability of these frameworks in systems supporting varying numbers of network slices.

The implementation study employs a Python-based simulation to model the IBFs and view functions (VFs), which represent the key MFs in the H-SliceMaster framework. Table IV states the simulation settings that incorporate randomized communication latency and bandwidth requirements between IBFs and VFs to reflect real-world network conditions. Depending on the communication protocol used between MFs, and without loss of generality, the bandwidth per signaling message requirement is assumed to be normally distributed with the stated parameters. The total bandwidth consumption in the system is determined by the bandwidth usage per MF (IBF or VF) and the message exchange rate per network slice. The total signaling bandwidth overhead per hour can be expressed as

$$\text{Total Bandwidth} = \sum_{\text{slice}=1}^{N_s} \sum_{\text{MF}=1}^{N_m} \sum_{\text{msg}=1}^{M_s} B_{\text{msg}}(\text{slice}, \text{MF}) \quad (1)$$

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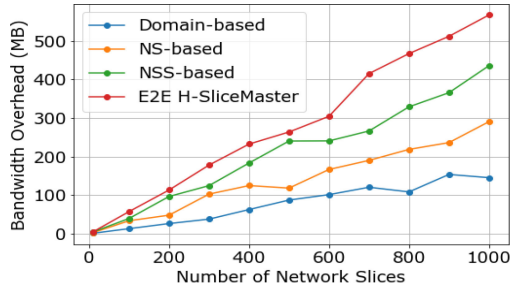


Fig. 5. Bandwidth signaling overhead of H-SliceMaster frameworks.

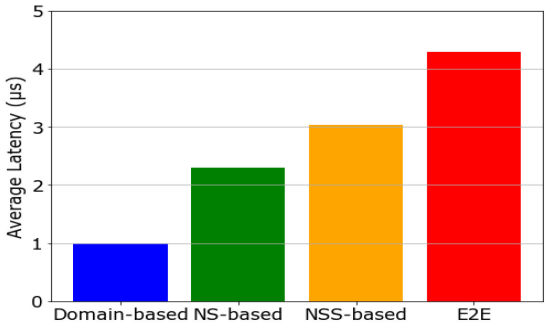


Fig. 6. Average latency of H-SliceMaster frameworks.

where N_s is the total number of network slices, M_s is the number of messages exchanged per MF per hour, MF is the Number of MFs, and B is the bandwidth consumption of message msg, network slice slice, and MF MF.

The latency between MFs is characterized by the geographic separation between their deployment locations. Drawing from empirical measurements reported in [49], which shows that the one-way latency is around 0.95 ms per 100 kilometers. In our analysis, we presumptively consider MFs within the H-SliceMaster architecture as being deployed in close geographic proximity, specifically within a 1-kilometer radius. Moreover, we quantify the average total latency experienced by signaling messages through an aggregation approach that includes the latency between the MFs and the E2E MO entity and the latency experienced during inter-MF communications. For example, considering the NSS-based framework, signaling messages traversing from the NSSVF to the E2E MO entity and the latency incurred during signaling between the NSSVF and the IBF are considered. It is noteworthy that all MO frameworks inherently experience latency between the E2E MO entity and the 3GPP MFs. Consequently, this baseline latency component is excluded from our comparative analytical framework.

Lastly, to study the scalability of the system, the signaling overhead is computed for different numbers of deployed network slices. For each system with a specific number of deployed network slices, the simulation computes the average MO decision latency and the total bandwidth overhead required for signaling exchanges.

Fig. 5 illustrates the bandwidth overhead of the H-SliceMaster frameworks for varying numbers of network slices, ranging from 10 to 1000. As anticipated, the figure demonstrates that as the number of network slices and MFs increase, the total bandwidth overhead scales proportionally. Despite this increase, the overhead remains within an acceptable range.

For instance, in the E2E H-SliceMaster framework, which instantiates eight MFs, the total bandwidth consumption reaches approximately 100 MB per hour when managing 200 network slices. This results in an average overhead of 0.5 MB per slice per hour. If deployed within a 5G core network with a 1 Gb/s capacity, the corresponding bandwidth overhead is 0.2 Mb/s, which represents 0.02% of total available bandwidth. This percentage indicates that the overhead is not a significant burden for high-speed networks such as 5G and cloud-based infrastructures. Similarly, in enterprise networks with a 100 Mb/s capacity, the most privacy-preserving E2E H-SliceMaster framework incurs a bandwidth overhead of 0.2%. This suggests that while the overhead remains moderate, it is still within tolerable limits for enterprise-level deployments supporting 200 network slices.

Fig. 6 plots the average latency experienced by signaling messages in the H-SliceMaster frameworks. The E2E H-SliceMaster framework demonstrates the most pronounced latency overhead, with signaling messages experiencing an average delay of approximately 4 microseconds when communicating with the core network functions. In contrast, the domain-based framework exhibits significantly reduced latency, averaging around 1 microsecond for comparable message transmissions.

This observed latency differential represents a tradeoff in privacy preservation and latency overhead. The enhanced privacy preservation mechanisms of the E2E H-SliceMaster framework require additional MFs that increase signaling delay. It is worth noting that practical framework implementations can enable direct communication channels between system domains and core network functions that are exclusive for critical emergency and event-driven communications.

V. Proof of Concept Study: Dynamic Intent-Oriented NS Embedding

In the CSMF of the H-SliceMaster, an intent is translated into a network slice template that includes an SFC and specific QoS requirements. The PoC focuses on the E2E resource embedding of intent-driven network slices, taking into account the RAN, edge, and cloud domains. This includes optimizing PPVs in the NSMF, which acts as a centralized E2E MO entity, and generating PPLs within the system domains. Additionally, the study accounts for variations in service demand and time-varying network conditions, such as resource utilization. However, scheduling is not considered, and it is assumed that the VNF chain is dependent, meaning the order of VNFs remains fixed.

A. System Model

The system model considers an IoT system employing the H-SliceMaster to manage and orchestrate several NSs. A summary of the mathematical notations is presented in Table V. Each of these NSs denoted as i, is instantiated for

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TABLE V
MATHEMATICAL NOTATION AND SYMBOLS

Symbol	Description
System Model Parameters	
i	Network slice index
T_i	Duration of network slice i
T_o	System running duration
Q_i	QoS requirement for slice i
PN_i	Service degradation penalty for slice i
RAN Model Parameters	
r	RAN system index
$D_{r,i}(t)$	Number of devices in RAN r for slice i at time t
$TD_r(t)$	Total number of devices in RAN r
$X_i^r(t)$	Traffic rate at RAN r for slice i
BW	Total wireless system bandwidth
N	Total number of resource blocks
$R_{NI}(t)$	RB slicing matrix
$R(n,i)(t)$	Binary entry indicating RB allocation
$RR_i(t)$	Number of RBs allocated to slice i
$Th_i(t)$	Throughput of slice i
B	Number of sub-channels
F	Number of sub-frames
b	Bandwidth of sub-channel
δt	Sub-frame length
TF	Frame interval
$\lambda(k_i)$	Average arrival rate during period k_i
$eRAN(k_i)$	Average transmission delay in RAN
Edge and Cloud Network Parameters	
P	Set of physical nodes
L	Set of directed links
$l_{v,z}$	Physical link between nodes v and z
$J_{g,h}$	Set of shortest paths between nodes g and h
$P_{CPU}(p_v)$	Processing capacity of node p_v
$L_{BW}(l_{v,z})$	Bandwidth capacity of link $l_{v,z}$
$U_v(t)$	Available processing capacity of node v
$U_{v,z}(t)$	Available bandwidth of link (v,z)
F_i	Set of VNFs for slice i
$M_i(t)$	Mapped forwarding graph
$A_{w,h}^i(t)$	Processing resources allocation
$R_{f_{i,w}}$	Output/input data rate ratio of VNF $f_{i,w}$
$P_{f_{i,w}}$	Processing demand per bandwidth of VNF $f_{i,w}$
QoS and Performance Metrics	
$Monitor(k_i)$	QoS monitoring function
$E2E_{QoS}(k_i)$	End-to-end QoS estimate
$eP(k_i)$	Processing delay
$eT(k_i)$	Transmission delay
Cost Model Parameters	
PR_{RB}	Cost rate of resource block
PR_{CPU}	Cost rate of CPU resources
PR_{BW}	Cost rate of bandwidth
$Cost_{E2E_i}(t)$	Total cost of deploying solution for slice i
$Cost_Q$	QoS violation cost
DRL Parameters	
s_t	State at time t
a_t	Action at time t
$\bar{D}_i(t)$	Average service demand
$\bar{R}_i(t)$	Average allocated resources
t_n	Discrete time
$Act(t)$	Action space
SC	Number of scaling options
γ	Discount factor
$Q_\pi(s,a)$	Action-value function
π	Policy

a duration T^i and serves a specific intent that is associated with a QoS level Q_i and a penalty for service degradation, represented as PN_i. We assume the system has been running for a duration of T_o. Therefore, the usage period of NS i corresponds to $T_o < t < T_o + T^i$. Moreover, the network data is analyzed and updated during the period T_o. The network environment accommodates a total of I instantiated network slices, where $i \in \{0, I\}$. In the subsequent sections, we delve into the specifics of system domain models.

1) RAN Model: The IoT network consists of RAN systems denoted by r, where $r = \{0, 1, \ldots, TR\}$, and TR is the total number of RAN systems in the IoT network. Each r serves a coverage area deploying I NSs. Each NS has a total of $D_{r,i}(t)$ devices. The total number of devices served by a RAN system is $TD^r(t) = \sum_i D_{r,i}(t)$. The set of devices belonging to NS i and system r is denoted by $d_{r,i} = \{0, 1, 2, \ldots, D_{r,i}(t)\}$. Furthermore, the traffic originating from the device domain utilizes the resources of the corresponding network slice i. To capture the time-varying traffic of each NS, $X_r^i(t)$ presents the traffic rate at RAN r for NS i at time instant t in bits per sec. We assume that RAN systems have identical capabilities and resources, hence we omit the dependency on r in the rest of the illustration.

1) *Physical Resources:* To fulfill the service requirements, it is essential to have the ability to allocate radio resources flexibly between network slices. Leveraging the inherent flexibility of the physical layer, the time and frequency resources within the RAN can be dynamically divided into physical resource blocks (RBs). The overall wireless system bandwidth is denoted as BW. The frequency spectrum is uniformly segmented into B subchannels, each with a bandwidth of b. Similarly, within the time domain, scheduling frames can be partitioned into F subframes, each with a length of $\bar{t}$. This division allows us to establish the minimum granularity of time-frequency resources, represented by the RB. A set of these time-frequency RBs is denoted by $n = 1, 2, 3, \ldots, N$, and $N = B.F$, which is the product of the total number of subchannels and subframes. Therefore, each r supports N RBs at each frame interval $TF = F.\bar{t}$.
2) *RAN Slicing Model:* We define matrix $\mathbf{R}_{N\times I}(t)$ to represent the RB slicing result of a RAN system. $\mathbf{R}_{N\times I}(t)$ has binary entries $R(n,i)(t)$ in which a value of 1 indicates that the RB n is allocated to slice i, otherwise a value of 0 is given, illustrated by the following:

$$R(n,i)(t) = \begin{cases} 1, & RB\ n \text{ allocated to NS}_i \\ 0, & \text{Otherwise.} \end{cases} \tag{2}$$

The following constraint ensures RBs cannot be shared within NSs.
Moreover, the total number of RBs allocated to an NS i is

$$RR_i(t) = \sum_n R(n,i)(t). \tag{3}$$

3) *QoS Model:* The throughput of NS i can be found according to

$$Th^i(t) = RR_i(t) * R \tag{4}$$

where R represents the data rate of one RB. We assume the processing delay experienced by the traffic to be neglected. Moreover, the traffic of a RAN system is modeled as an $M/M/1$ queuing model. To reflect the varying network traffic $X^i(t)$, we define $\lambda(k^i)$ to measure the average arrival rate during an observation period

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K, where $k^i = \{0, 1, 2, \ldots, |k^i|\}$ is a time index that runs during the instantiation period of NS i. Hence, the following holds, $T^i = |k^i|.K$. Moreover, $\overline{\text{th}}^i(k^i)$ is the average NS throughput in the interval k^i. The average transmission delay of NS i can be expressed as

$$\text{eRAN}\left(k^i\right) = \frac{1}{\overline{\text{th}}\left(k^i\right) - \lambda\left(k^i\right)}. \tag{5}$$

2) Edge and Cloud Network Model: The edge and cloud network consists of VNF-capable nodes and bi-directed links. An NS request can be deployed on the edge, on the cloud, or both networks.

1) *Physical Network:* The edge and cloud network is modeled as a directed graph $G = (P, L)$, where P and L are the sets of $|P|$ physical nodes and $|L|$ directed links, respectively. The link $l_{\upsilon,z} \in L$ represents the direct physical link connecting physical nodes p_υ to p_z $\forall p_\upsilon, p_z \in P$. Let $J_{g,h} = \{j^1_{g,h}, \ldots, j^s_{g,h}\}$ be the set of shortest paths between physical nodes p_g to p_h. Each node $p_\upsilon \in P$ has a processing capacity $P_{\text{CPU}}(p_\upsilon)$ and each link $l_{\upsilon,z} \in L$ has a bandwidth capacity $L_{BW}(l_{\upsilon,z})$. To capture the time-varying resource utilization of the system, we define $U_\upsilon(t)$ and $U_{\upsilon,z}(t)$ to be the amount of available processing capacity of node p_υ and the available bandwidth of link $l_{\upsilon,z}$, respectively.
2) *NS Request:* For an NS i the set $F_i = \{f_{i,1}, f_{i,2}, \ldots, f_{i,|F_i|}\}$ determines the requested SFC consisting of a total of $|F_i|$ VNFs. The order of running the VNF chain is already known. Let $R_{f_{i,w}}$ be the ratio of the outgoing data rate to the incoming data rate of VNF $f_{i,w} \in F_i$. Moreover, let $P_{f_{i,w}}$ denote the processing resource demand per bandwidth of VNF $f_{i,w} \in F_i$ measured in cycles/bit. We assume those request parameters remain unchanged during the NS instantiation period, e.g., $T_o < t < T_o + T_i$.
3) *Resource Slicing:* Let the mapped forwarding graph $M_i(t)$ of a NS i be an ordered list of $|M_i|$ distinct elements. Since the order of the VNFs is known and each VNF has one active deployment in the network, the number of elements in $M_i(t)$ will equal the number of VNFs $|F_i|$. Let the wth element of $M_i(t)$ be $\sigma^{s,g,h}_{i,w}(t) = <A^i_{w,h}(t), j^s_{g,h}>$, which contains two values. The first is a mapping of VNFs into physical nodes, according to

$$A^i_{w,h}(t) = \Gamma^i_{w,h}.A_h(t) \tag{6}$$

where the indicator function $\Gamma^i_{w,h}$ is

$$\Gamma^i_{w,h} = \begin{cases} 1, & \text{if } f_{i,w} \text{ is deployed on } p_h \\ 0, & \text{Otherwise.} \end{cases} \tag{7}$$

Hence, $A^i_{w,h}(t)$ represents the amount of processing resources (cycles/sec) allocated by $p_h \in P$ to $f_{i,w} \in F_i$. The second value in $M_i(t)$ is $j^s_{g,h} \in J_{g,h}$. In addition, the function $MAP_y(\sigma^{s,g,h}_{i,w})$ returns the component of each element as follows:

$$\text{MAP}_y\left(\sigma^{s,g,h}_{i,w}(t)\right) = \begin{cases} A^i_{w,h}(t), & \text{if } y = 1 \\ j^s_{g,h}, & \text{if } y = 2. \end{cases} \tag{8}$$

Therefore, the elements $\sigma^{s,g,h}_{i,w}(t)$ indicate that for an SFC mapping graph $M_i(t)$, the VNF $f_{i,w}$ is deployed on node p_h which allocates processing resources $A_h(t)$ and connected to the previous physical node through the shortest path $j^s_{g,h}$. Moreover, the first element $w = 1$ defines the ingress node which connects the RAN system to the first VNF function $f_{i,1}$. Additionally, we define the sets $P_{M_i} \subseteq P$ and $J_{M_i} \subseteq J$ of all physical nodes and paths utilized by the mapping graph M_i, respectively.

4) *QoS Model:* The QoS model considers the average delay experienced through the processing and transmission of NS traffic. Let $X^i_w(t)$ be the incoming traffic rate of VNF $f_{i,w}$. We define $\lambda^w(k^i)$ to be the average of $X^i_w(t)$ in a time interval K. We model the processing of VNFs by an M/M/1 queuing model. Hence, the average processing delay of NS i can be expressed as

$$eP\left(k^i\right) = \sum_{h|h\in P_{M_i}} \sum_{w|w\in F_i} \frac{1}{\frac{A^i_{w,h}\left(k^i\right)}{P_{f_{i,w}}} - \lambda^w\left(k^i\right)} \tag{9}$$

where $A^i_{w,h}(k^i)$ is the average processing capacity allocated to NS i during the time interval of k^i. Similarly, the transmission delay can be expressed by

$$eT\left(k^i\right) = \sum_{j^s_{g,h}|j^s_{g,h}\in J_{M_i}} \sum_{\upsilon,z|l_{\upsilon,z}\in j^s_{g,h}} \frac{1}{L_{BW}\left(l_{\upsilon,z}\right) - \lambda^g\left(k^i\right)} \tag{10}$$

where $\lambda^g(k^i)$ is the average resource demand from node g.

B. Problem Formulation

According to the aforementioned system model, we define the E2E dynamic NS resource embedding problem to be the problem of finding the RAN slicing solution, denoted by $\mathbf{R}_{\text{NXI}}(t)$ and the core network mapped forwarding graph denoted by $M_i(t)$, for the NS lifecycle period $T_o < t < T_i$. We define E2E_i to be the set of candidate E2E resource embedding solutions $|\text{E2E}_i|$, in which each element of E2E_i has two values, $\sigma^{\text{E2E}}_{i,c} = <\mathbf{R}_{\text{NXI}}(t), M_i(t)>$.

1) *Resource Constraints:* Each candidate solution in E2E_i should ensure the following constraint in the RAN domain:

$$\sum_n \sum_i R(n,i)(t) \leq N \ \forall t > T_o \tag{11}$$

which ensures the total number of allocated RBs does not exceed the total number of RBs in the system. Moreover, for a candidate-mapped graph $M_i(t)$ solution in E2E^i the following constraint ensures that the allocated processing resources for all NSs do not exceed the available resources:

$$\begin{aligned} &\sum_{\forall\{i,w\}|f_{i,w}\in V_{M_i}} P_{f_{i,w}}.X^i_w(t) \leq U_\upsilon(t) \\ &\forall p_\upsilon \in P_{M_i} \ \forall t > T_o \\ &\sum_{\forall\{i,w\}|f_{i,w}\in V_{M_i}} R_{f_{i,w}}.X^i_w(t) \leq U_{\upsilon,z}(t) \end{aligned} \tag{12}$$

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$$\forall p_\upsilon \quad \text{and} \quad p_z \in P_{M_i} \ \forall \{\upsilon, z\} \in j^s_{g,h} \ \forall t > T_o \tag{13}$$

where V_{M_i} is the set of VNFs that are deployed at node p_υ.

2) *Cost Model*: The cost model of the dynamic NS resource embedding problem considers two components, the first is the cost of allocated resources, including the wireless communication links, bandwidth, and processing resources. In addition, the model includes a penalty cost associated with service degradation. Specifically, NSs come with a specified QoS level, in this work, we consider the E2E delay threshold identified as Q^i, and a penalty for service degradation denoted by P^i.
The resource embedding cost is expressed as

$$\begin{aligned}\text{Cost}^R_{\text{E2E}_i}(t) = {} & RR_i(t).PR^{RB} \\ & + \sum_{\forall h \in P_{M_i}} \sum_{\forall w \in F_i} A^i_{w,h}(t).PR^{\text{CPU}} \\ & + \sum_{\forall \upsilon, z \in J_{M_i}} L_{BW}\left(l_{\upsilon,z}\right).PR^{BW}\end{aligned} \tag{14}$$

where the first term finds the cost of RBs, the second term finds the cost of processing resources and the third term finds the cost of bandwidth. The values PR^{BW}, PR^{CPU}, PR^{BW} are the monetary cost rate of a unit of RB, CPU, and bandwidth resources, respectively.
The system also considers the cost of violating the QoS constraints of the intent as follows:

$$\text{Monitor}\left(k^i\right) = \begin{cases} 1, & \text{if } \text{E2E}_{\text{QoS}}\left(k^i\right) > Q^i \\ 0, & \text{Otherwise} \end{cases} \tag{15}$$

where the function Monitor(k^i) monitors the E2E QoS estimate $\text{E2E}_{\text{QoS}}(k^i)$ of NS i during an observation period k^i. The Monitor(k^i) function has a value of 1 if the observed E2E QoS metric exceeds the requirement of the NS. The E2E QoS metric considered in this study is the delay estimated by

$$\text{E2E}_{\text{QoS}}\left(k^i\right) = \text{eRAN}\left(k^i\right) + eP\left(k^i\right) + eT\left(k^i\right). \tag{16}$$

Accordingly, the QoS violation cost can be expressed by

$$\text{Cost}^Q_{\text{E2E}_i}(t) = \text{Monitor}\left(k^i\right).P^i, \ \ k^i = \lfloor \frac{t}{K} \rfloor. \tag{17}$$

The total cost of deploying an mapped solution E2E_i is

$$\text{Cost}_{\text{E2E}_i}(t) = \text{Cost}^R_{\text{E2E}_i}(t) + \text{Cost}^Q_{\text{E2E}_i}(t). \tag{18}$$

3) *Optimization Objective*: Considering the above requirements, the objective of the dynamic NS resource embedding problem becomes

$$\begin{aligned}\min \quad & : OBj = \int_{T_o < t} \sum_{i \in I} \text{Cost}_{\text{E2E}_i}(t)\, dt \\ \text{s.t.} \quad & \text{Constrains (11)}-\text{(13)}.\end{aligned} \tag{19}$$

C. Proposed Promise and Price Algorithm

The implementation of the PPNO principle encompasses two primary tasks: 1) the generation of PPTs within individual system domains and 2) the optimization of E2E PPVs at the NSMF. This section proposes a specific methodology for realizing the PPNO principle while acknowledging that the principle itself represents a general framework that can accommodate various implementation approaches.

1) Intradomain Promise and Price Table: The proposed PPNO algorithm incorporates the unique characteristics of IoT traffic into the design of the PPT. Given the diverse traffic patterns observed in IoT networks [46], [50], the algorithm captures these time-varying characteristics by developing resource autoscaling policies tailored to specific stochastic IoT traffic profiles. These policies are designed to meet specific QoS requirements while minimizing resource allocation, considering the dynamic fluctuations in traffic patterns.

1) *Promise:* A promise is defined as a combination of a traffic stochastic profile (TSP), a corresponding QoS threshold such as a delay threshold, and a promise guarantee value.
2) *Policy:* To achieve optimal resource autoscaling policies for various TSPs and QoS thresholds, we propose the implementation of deep Q-networks (DQNs). DQN, a reinforcement learning technique, is particularly well-suited for this task due to its ability to learn complex, nonlinear relationships between states, actions, and rewards.
The resource scaling problem can be formulated as a Markov decision process (MDP), described by $<S, A, C, P, \gamma>$, representing the: state space, action set, cost, transition probability, and discount factor. In the following, the details of the MDP elements are described.
   a) *State Space:* The state information includes the average demand in a past window, the number of allocated resources, and the time information. The system collects the average service demand in a monitoring window $\bar{D}^i(t)$ and the number of allocated resources $\bar{R}^i(t)$. A discrete-time t_n is added to capture temporal traffic fluctuation information. Then, the state s_t of a network slice i at time instant t can be expressed as $s_t = \{\bar{D}^i(t), \bar{A}^i(t), t_n\}$.
   b) *Action Set:* The resource allocation policy decides on the scaling of Virtual Resource vR units. Those vR units represent radio resource blocks in the RAN network and virtual computing resources in the edge and cloud networks. Therefore, the algorithm has a total of SC scaling options, scaling up by 1 and up to SC vR units, scaling down by 1, or up to SC vR units, or not scaling the resources. Hence, the action space is denoted by $\text{Act}(t) \in \{0, 1, \ldots, (2 * SC)\}$, where each numeric value of an action a corresponds to a scaling option.
   c) *Reward:* The reward associated with an action Act(t) is related to the observed delay in the system, defined as

$$\text{Reward}(t) = \begin{cases} \infty, & \text{if } \overline{\text{Delay}}(t) < D_{\text{th}} \\ -\text{Cost}^R, & \text{Otherwise} \end{cases} \tag{20}$$

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where $\overline{\text{Delay}}(t)$ is the average observed delay in the system, found using (4) for RAN systems and (9) for edge and cloud systems. Moreover, Cost^R represents the system costs which is found by the first term in (13) for RAN systems and the summation of the second and third terms of (13) for the edge and cloud systems. Lastly, D_{th} is a delay threshold that the policy tries to achieve.

At the time t, the total reward is defined as the cumulative discounted reward, as follows:

$$\begin{aligned}\text{TotalReward}(t) &= \text{Reward}(t) + \gamma\text{Reward}(t+1) + \cdots \\ &= \sum_{k=0}^{\infty} \gamma^k \text{Reward}(t+k) \end{aligned} \tag{21}$$

where $\gamma \in (0, 1)$ is a discount factor for future reward in the cumulative reward function. Hence, the objective becomes to find the policy (π) that

$$\max_{\pi} \ \mathbf{E}\left(\sum_{t=0}^{\infty} \gamma^t \text{Reward}(t)\right). \tag{22}$$

d) *Deep Q Networks:* The MDP model does not analyze the transition probabilities of the state space due to the unknown traffic variation. DRL addresses this problem by utilizing experience in the environment. Specifically, an agent in DRL interacts with the environment repeatedly. Those experiences are used to learn an optimal decision policy π that minimizes the long-term cumulative reward as defined by (20). One of the most popular DRL algorithms is Q-learning. Q-learning is an algorithm that utilizes the experience of the agent's interaction with the environment to learn the action-value function $Q^{\pi}(s, a)$

$$Q^{\pi}(s_t, a_t) = \mathbf{E}\left(\sum_{t=0}^{T} \gamma^t \text{Reward}(t) | s_t, a_t, \pi\right). \tag{23}$$

Consequently, the optimal mapping Q^* is obtained by $Q^*(s_t, a_t) = \max_{\pi}(Q^{\pi}(s_t, a_t))$ and satisfies the Bellman equation as follows:

$$\begin{aligned} Q^*(s_t, a_t) &= \text{Reward}(s_t, a_t) \\ &\quad + \max_{a'} \ \gamma \mathbf{E}\left[Q^*(s_{t+1}, a_{t+1})\right]. \end{aligned} \tag{24}$$

Therefore, by knowing the optimal Q-function Q^*, the optimal policy becomes: $\pi(s) = \arg\max_{a'} Q^*(s_t, a_t)$.

3) *Price:* The price associated with each promise is the anticipated cumulative cost of allocated resources for the intent's activation period, as follows:

$$\int_{T_o}^{T_i} PR^R * R(t)\, dt \tag{25}$$

which reflects the cost of the total resources to be allocated to a specific TSP and a QoS threshold using the optimal policy obtained by (24), during the NS activation period. Note that depending on the intradomain system, the value of PR^R is equivalent to either PR^{BW}, PR^{CPU}, PR^{BW} which are the monetary cost rate of a unit of RB, CPU, and bandwidth resources, respectively.

4) *Promise Guarantee:* The promise guarantee component of the PPT represents a probabilistic measure of a node's capability to sustain its QoS commitments over an extended operational period. This metric is derived through statistical analysis of two time series: a) the resource availability function $A(t)$ and b) the resource demand function $D(t)$, where $t \in [T_o, T_o + T_i]$ represents the operational timeline of the network slice.

Let $s(t) = A(t) - D(t)$ denotes the resource surplus function. The promise guarantee value PG for a specific delay threshold D_{Th} is then defined as

$$PG = P(s(t) \geq 0) \ = 1 - F_s(0) \tag{26}$$

where $F_s(\cdot)$ represents the empirical cumulative distribution function (ECDF) of $s(t)$. This formulation provides a probabilistic measure of a node's ability to maintain sufficient resources to support the designated delay threshold D_{Th}. Moreover, this formulation reveals that PG values demonstrate a strong positive correlation with both delay performance and resource availability. Higher PG values indicate an enhanced probability of maintaining the prescribed delay thresholds, attributed to more consistent resource availability. This metric's particular strength lies in its incorporation of temporal dynamics in both resource availability and demand patterns, providing a more realistic assessment of node capabilities compared to static resource analysis.

2) End-to-End Policy Optimization: E2E policy optimization is carried out at the NSMF and the NSSMF, if it consists of multiple system domains. These functions receive detailed system snapshots, which include the PPL. Based on the specific optimization objectives, the E2E algorithm may prioritize system performance, by emphasizing guarantee values, or resource costs by considering price values. In this section, we design a dynamic NS deployment algorithm for optimizing the E2E policy. We focus on the delay performance aspect, therefore, the algorithm maximizes guarantee values while ensuring E2E delay constraints. Two main functions are designed, Algorithm 1 shows the dynamic NS deployment which allocates delay budgets to the VNFs and RAN system, and Algorithm 2 which optimizes the VNF embedding on the physical network.

The E2E performance function is defined by the following:

$$\text{Performance} = \sum_{s=1}^{S} w_{\text{QoS}} \cdot PG^s + w_{\text{cost}} \cdot \frac{1}{\text{Price}^s} \tag{27}$$

where w_{QoS} and w_{cost} are values less than one capturing the importance of considering PG and price, respectively. Also, s is a PV index where the E2E MO entity receives a total of S PVs from underlying system domains. We note that this formulation is equivalent to the E2E optimization objective in (18). Nevertheless, the Performance equation allows different NSs, e.g., service providers, to prioritize either delay performance or the cost of resources.

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Algorithm 1 Dynamic E2E NS Deployment

Input: N: Number of VNFs, M: Number of servers, D_{req}: Total E2E delay requirement, P: Delay-guarantee and price values for each server, P_{RAN}: RAN delay-guarantee and price values, w_{QoS}, and w_{cost}
Output: Optimal VNF placement, selected RAN configuration, E2E delay, and guarantee values
Initialize $\alpha_{\text{init}} \leftarrow 0.9$, $\alpha_{\text{step}} \leftarrow 0.05$, $\alpha \leftarrow \alpha_{\text{init}}$
Initialize *best_alpha* $\leftarrow 0$, *best_e2e_delay* $\leftarrow \infty$, *best_PG* $\leftarrow -\infty$, *best_ran* $\leftarrow -1$, *best_price* $\leftarrow -\infty$
while $\alpha \geq 0.5$ **do**
    $D_{\text{vnf}} \leftarrow \alpha \times D_{\text{req}}$ ▷ Allocate delay to VNF chain
    (*vnfs_placement*) $\leftarrow$ *FindOptimalPlacement*$(N, M, P, w_{QoS}, w_{cost}, D_{\text{vnf}})$
    if *vnfs_placement* is not None **then**
        $D_{\text{RAN}} \leftarrow D_{\text{req}} - D_{\text{vnf}}$
        for each $p_{\text{ran}} \in P_{\text{RAN}}$ **do**
            Performance$(w_{QoS}, w_{cost}) \leftarrow P_{\text{RAN}}[p_{\text{ran}}]$
            if $p_{\text{ran}} \leq D_{\text{RAN}}$ and *Performance*$[p_{ran}] >$ *best*$_{Performance}$ **then**
                best_ran $\leftarrow P_{\text{RAN}}[p_{\text{ran}}]$
                best_alpha $\leftarrow \alpha$
                best_e2e_delay $\leftarrow D_{\text{vnf}} + p_{\text{ran}}$
            end if
        end for
        break
    else
        $\alpha \leftarrow \alpha - \alpha_{\text{step}}$
    end if
end while
if *vnfs_placement* is not None **then**
    Return: *vnfs_placement*, *best_ran*, *best_e2e_delay*, *best_price*, *best_PG*
else
    Return: No feasible solution found
end if

Algorithm 1 takes several inputs: the number of VNFs to be placed N, the number of available servers M, a 2-D array P representing the promised delay and guarantee pairs for each server, the total E2E delay requirement D_{req}, a 2-D array of promised delay and guarantees for the RAN system P_{RAN}, and the w_{QoS} and w_{Cost} values. The algorithm initializes key parameters: α_{init}, the initial proportion of the total delay allocated to VNFs, set to 0.9, and α_{step}, the step size for reducing the VNF delay proportion, set to 0.05.

The algorithm employs an adaptive approach. It starts by allocating 90% of the total delay to VNFs ($\alpha = 0.9$) and iteratively reduces α by α_{step} if no feasible solution is found. This process continues until a solution is found or α becomes less than 0.5. For each iteration, the algorithm calculates the VNF delay budget based on the current α, calls FindOptimalPlacement to find a VNF placement solution, selects a compatible RAN option if a solution is found, and updates the best solution if the current solution has a lower total cost.

Algorithm 2 SFC Placement Algorithm

Input: N: Number of VNFs, M: Number of servers, P: Delay-guarantee pairs for each server, D_{vnf}: Total delay constraint for SFC, w_{QoS}, w_{Cost}
Output: Optimal VNF placement and maximum guarantee value
total_delay_constraint $\leftarrow \lfloor D_{\text{vnf}} \rfloor$
Initialize $dp[N+1][total_delay_constraint+1] \leftarrow 0.0$
Initialize $choices[N+1][total_delay_constraint+1] \leftarrow$ None
Initialize $used_nodes[N+1][total_delay_constraint+1] \leftarrow \emptyset$
for *vnf* $= 1$ to N **do**
    for *each delay_budget* **do**
        for each server *node_id* $= 1$ to M **do**
            for each (*delay*, *guarantee*) in $P[node_id]$ **do**
                if *delay* $\leq$ *delay_budget and* *node_id* $\notin$ *used_nodes* **then**
                    new_value $\leftarrow$ $dp[vnf-1][delay_budget - delay] + Performance(w_{QoS}, w_{cost})$
                    if *new_value* $>$ $dp[vnf][delay_budget]$ **then**
                        $dp[vnf][delay_budget] \leftarrow$ *new_value*
                        choices $\leftarrow$ (*node_id*, *delay*)
                        used_nodes $\leftarrow$ *node_id*
                    end if
                end if
            end for
        end for
    end for
end for
max_*guarantee* $\leftarrow dp[N][total_delay_constraint]$
Initialize empty list *vnfs_placement*
current_delay_budget $\leftarrow$ *total_delay_constraint*
for *vnf* $= N$ to 1 **do**
    (*node_id*, *delay*) $\leftarrow$ *choices*
    Append (*vnf*, *node_id*, *delay*) to *vnfs_placement*
    current_delay_budget $\leftarrow$ *current_delay_budget* $-$ *delay*
end for
Return: *vnfs_placement*, max_*guarantee*

The FindOptimalPlacement function uses dynamic programming to find the optimal VNF placement for a given VNF delay budget as illustrated by Algorithm 2. A 2-D dynamic programming table (*dp*) is created and initialized with zeros, indicating that no VNFs are placed initially. A choice table (choices) is created to record the node ID and delay threshold used to achieve the maximum guarantee value for each state. Lastly, a used nodes table ($\text{used}_{\text{nodes}}$) is created to track the nodes used for each VNF.

The algorithm iterates over each VNF and each possible delay budget in descending order to avoid overwriting values that are needed for subsequent calculations. For each VNF,

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and each delay budget, it checks all available servers (nodes) and their corresponding delay thresholds and guarantee values. If a node and delay threshold can be used (i.e., the node has not been used in the current configuration), it computes a new guarantee value by adding the guarantee associated with the current node and delay threshold to the maximum guarantee value achievable with one less VNF and the remaining delay budget. The *dp* table is updated with the maximum guarantee value obtained, and the choices table is updated to record the node and delay threshold used.

After populating the *dp* table, the optimal guarantee value is found to represent the maximum guarantee value achievable with all N VNFs and the total delay constraint D_{vnf}. The solution is reconstructed by backtracking through the choices table. Starting from the final state, the algorithm traces back the nodes and delay thresholds used to achieve the maximum guarantee value and collects them into the $\text{vnfs}_{\text{placement}}$. The list $\text{vnfs}_{\text{placement}}$ is reversed to present the VNF placements in the correct order, showing which node and promised delay threshold were selected for each VNF.

VI. Simulation Study

This section presents a comprehensive performance evaluation of the proposed H-SliceMaster framework, with particular emphasis on the PPNO and SNC mechanisms. Initially, we characterize diverse IoT services and their associated TSPs. Subsequently, we delineate the simulation environment, including system parameters and experimental settings. After that, we conduct a thorough comparative analysis of the PPNO principle against state-of-the-art approaches, specifically the branching dueling Q-network (BDQN) and greedy algorithms. Finally, we validate the efficacy of the SNC mechanism through a case study involving network operator intent implementation, demonstrating its capability to autonomously achieve desired system behaviors while maintaining domain privacy.

A. Traffic Stochastic Profiles

IoT traffic exhibits distinct application-dependent temporal characteristics, manifesting in diverse traffic fluctuation patterns and varying degrees of burstiness. We encapsulate these dynamic behavioral patterns within the concept of TSPs. This study examines three representative TSPs corresponding to distinct IoT service categories: smart city, eHealth, and smart factory, as illustrated in Fig. 7. To facilitate comprehensive evaluation, we leveraged the tiered Markov modulated stochastic process (TMMSP) framework to generate an extensive dataset comprising 90 different time-series sequences, each spanning 10 days. The dataset maintains a balanced representation across the three TSPs, with 30 sequences per profile. The TMMSP parameters employed for traffic generation are detailed in [46].

B. System Settings

For our experimental evaluation, we implemented a comprehensive simulation environment using Python with the parameters summarized in Table VI that aligns with closely

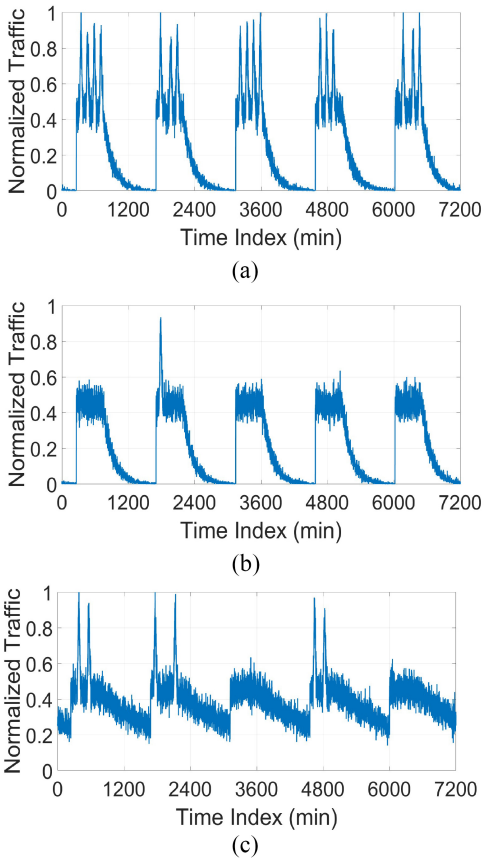


Fig. 7. Normalized traffic volume of (a) Smart city, (b) Smart factory, and (c) eHealth services.

TABLE VI
Simulation Parameters

Variable	Values / Range
$G = (P, L)$	(37, 64)
$P_{CPU}(p_v)$	Core: [1, 5] GHz [45]; Edge: [1, 3] GHz
$L_{BW}(l_{v,z})$	[100, 1000] Mbps [45]
$\lvert F_i \rvert$	4 [45]
cpu_i	[10, 99] cycles/s
$R_{f_{i,w}}$	[0.5,1.5] [45]
$P_{f_{i,w}}$	0.5 [45]
N	10 RBs [11]

related work in the literature. The core network topology is based on the Nobel-Germany network from SNDlib [51], which represents a realistic backbone network infrastructure. This topology comprises 17 nodes, where each node is assigned computing resources randomly distributed between 2–5 GHz for core nodes and 0.5–2 GHz for edge nodes. The internode links are configured with bandwidths ranging from 100 Mb/s to 1 Gb/s. We extended this core infrastructure using NetworkX, a Python library for complex network analysis, to incorporate access and edge network segments of 30 and 20 nodes, respectively. This results in a three-tier hierarchical network architecture that closely mirrors real-world deployments. To create realistic load conditions and capture temporal variations in resource utilization, we simulated the concurrent operation of 90 network slices, equally distributed between the three service types, eHealth, smart factory, and smart city over

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TABLE VII
DQN HYPERPARAMETERS

Parameters	Description
The learning rate of Adam optimizer	0.002
Discount factor γ	0.97
Experience mini-batch size	100
Replay buffer size	10000
Exploration parameter	0.1 to 0.01

five days. This generated a comprehensive dataset of resource utilization patterns across different network nodes.

C. System Domain Promise and Price Table

The formulation and maintenance of the PPT within each system domain represents a critical component of the proposed framework, encompassing two fundamental aspects: 1) the establishment of robust resource allocation policies tailored to specific delay requirements and 2) the derivation of quantifiable promise guarantees. In this analysis, we examine the efficacy of the proposed D3QN-based policies in maintaining diverse delay thresholds for different TSPs. Furthermore, we analyze the resource availability patterns to derive meaningful promise guarantee metrics that accurately reflect each node's capability to sustain its committed service levels.

1) *D3QN Models:* To evaluate the robustness of our approach across different QoS requirements, we examined three distinct delay thresholds D_{th}: 2, 5, and 10 milliseconds, representing stringent, moderate, and relaxed delay constraints, respectively. For each combination of TSP and delay threshold, we trained a dedicated D3QN model, resulting in a total of nine distinct models. The D3QN architecture was implemented using PyTorch, with each network comprising four layers: an input layer, two fully connected hidden layers (each containing 100 neurons), and an output layer. We employed the Rectified Linear Unit (ReLU) activation function for the hidden layers to introduce nonlinearity. The Adam optimizer is used with mini-batch gradient descent, with hyperparameters detailed in Table VII. To ensure a thorough exploration of the state-action space while maintaining the exploitation of learned policies, we implemented an adaptive ϵ greedy strategy. The exploration parameter ϵ was initialized at 0.1 and decremented by 0.0002 per time step until reaching a minimum value of 0.02, ensuring a balance between exploration and exploitation throughout the training process.
2) *Convergence Performance:* The convergence analysis, illustrated in Fig. 8, reveals learning patterns across different delay thresholds and traffic profiles. All models demonstrate learning convergence within the first 30 episodes, characterized by steep improvements in reward values. We noticed that the convergence behaviors with more relaxed delay constraints—models trained with the 10 ms threshold exhibit larger reward values than those operating under stricter 2 ms requirements. This is attributed to the 10 ms delay threshold requiring less amount of allocated resources, e.g., CPU processing capacity, compared to the 2 ms delay threshold, which is reflected in the reward function defined in (19). Moreover, models trained on smart factory TSP consistently achieve slightly higher final reward values across all delay thresholds. This performance is attributed to the less bursty nature of smart factory traffic compared to the smart city and eHealth traffic, therefore requiring lower resources to achieve the same delay performance.
3) *Temporal Performance:* The temporal performance of the trained D3QN models were evaluated using a five-day validation dataset for each TSP. Fig. 9 presents the time-series analysis of packet delays, averaged over 10-min intervals, demonstrating the models' ability to maintain QoS constraints across varying traffic conditions. For all delay thresholds (D_{th}), the D3QN policies consistently maintain average packet delays below their respective constraints, except for a single bursty time interval in the eHealth traffic. The behavior is particularly evident in the most stringent case ($D_{\text{th}} = 2$ ms), where despite temporal fluctuations, the average delay remains strictly bounded within the [0, 2] ms interval. This robust performance validates the effectiveness of the learned policies in enforcing QoS thresholds. The delay performance exhibits distinct patterns across different TSPs. The eHealth traffic demonstrates stable delay patterns with minimal variance in the 2 and 5 ms thresholds, attributed to its low traffic volume fluctuation range. In contrast, Smart City and Smart Factory traffic show wider delay variations while maintaining threshold compliance, reflecting their more dynamic traffic patterns. This variation in delay behavior aligns with the inherent characteristics of each application domain, suggesting that the D3QN policies have successfully adapted to application-specific traffic dynamics.
4) *Cumulative Distribution Function (CDF) Performance:* The CDF analysis, presented in Fig. 10, provides a more granular view of the delay distribution, revealing several significant insights. All policies achieve CDF values exceeding 0.9 at their respective threshold points, indicating robust QoS fulfillment. Moreover, the maximum observed delay for any individual packet stays within a 2 ms margin of the specified threshold, demonstrating tight control over delay outliers. The CDF curves exhibit similar shapes across different TSPs. These results collectively demonstrate that the D3QN-based policies have successfully learned to adapt resource allocation decisions to maintain QoS requirements while accommodating the unique characteristics of different IoT traffic profiles.
5) *Promise Guarantee Performance:* The last component in the PPT is the promise guarantee which is found using the resource availability of each node and the resource demand of each D3QN policy, as discussed in Section III-C. Fig. 11 presents the promise guarantee distributions across the network topology, encompassing access, edge, and cloud nodes and for the different promises, e.g., D_{th} s. For the stringent delay threshold

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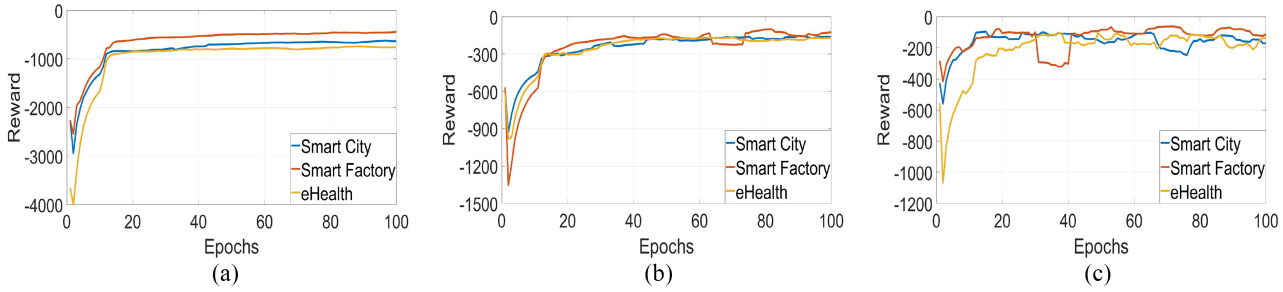


Fig. 8. Convergence performance for (a) $D_{\text{th}} = 2$ msec, (b) $D_{\text{th}} = 5$ msec, and (c) $D_{\text{th}} = 10$ msec.

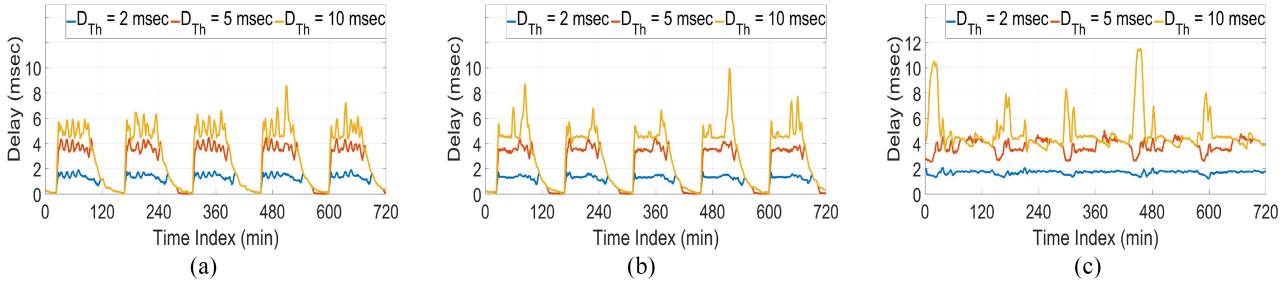


Fig. 9. Delay performance of (a) Smart city traffic, (b) Smart factory traffic, and (c) eHealth traffic.

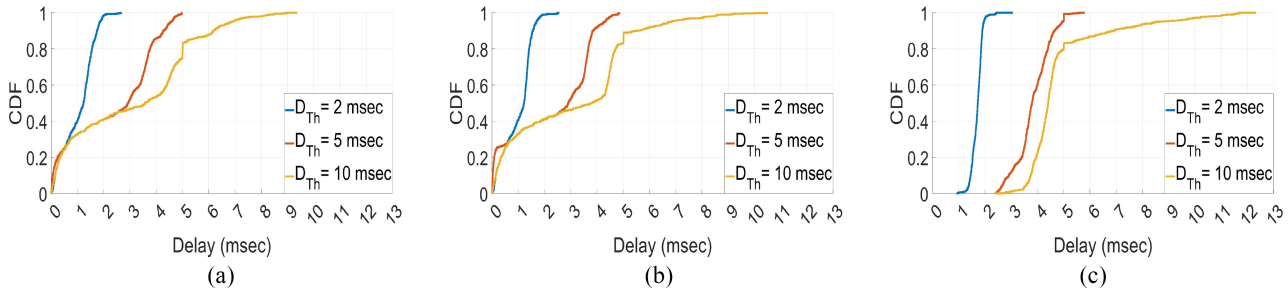


Fig. 10. CDF of delay performance for (a) Smart city traffic, (b) Smart factory traffic, and (c) eHealth traffic.

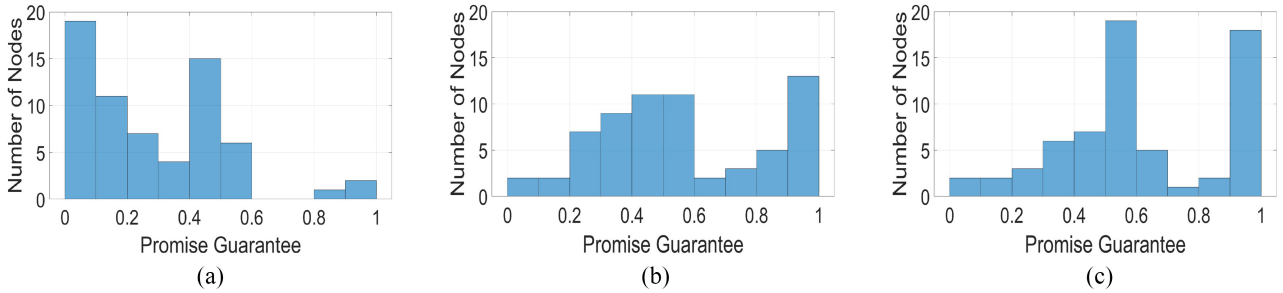


Fig. 11. Promise guarantee of (a) $D_{\text{th}} = 2$ ms, (b) $D_{\text{th}} = 5$ ms, and (c) $D_{\text{th}} = 10$ ms.

of D_{th}= 2 ms, the empirical results reveal that a significant proportion of network nodes (approximately 90%) exhibit PG values below 0.6. This statistical characterization carries profound operational implications: these nodes can guarantee sufficient resources to maintain the 2 ms delay threshold for less than 60% of the network slice's operational lifetime. Conversely, this indicates a probability exceeding 0.4 that these nodes will experience resource insufficiency at some point during their operational period.

Moreover, the promise guarantee figure reveals a notable correlation between delay threshold values and the distribution of promise guarantee values across network nodes. As the delay threshold D_{th} increases from 2 to 10 ms, we observe a systematic rightward shift in the promise guarantee distribution, indicating enhanced resource sufficiency probabilities across the network topology. This phenomenon is mathematically supported by the processing and transmission delay models presented in (8) and (9), which illustrate the inverse

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relationship between resource requirements and delay thresholds and explain why nodes exhibit higher promise guarantee values for larger D_{th} values. For instance, at $D_{\text{th}} = 10$ ms, a substantial proportion of nodes demonstrate promise guarantee values exceeding 0.8, indicating their capability to sustain service commitments for more than 80% of their operational timeline.

D. End-to-End Network Slice Deployment

The evaluation of the proposed PPNO principle encompasses a thorough analysis of E2E network slice deployment decisions across varying delay thresholds Q^i and a comparative assessment against state-of-the-art deployment algorithms. We deploy the E2E algorithm discussed in Section IV-B2 with the values of w_{QoS} and w_{cost} to be 1 and 0, respectively.

1) *Network Slice Deployment Decision:* The network slice deployment decisions, presented in Tables VIII–X, offer comprehensive insights into the PPNO principle's adaptation to different traffic profiles and QoS requirements. Each table presents the placement decisions in terms of selected delay thresholds (D_{th}) and their associated guarantee values across the service chain, encompassing the access domain and four VNFs, ultimately achieving specific E2E QoS objectives. For granular performance analysis, we examine deployment decisions across six distinct E2E delay thresholds Q^i ranging from 10 to 45 ms. The PPNO principle distributes delay budgets across network functions. For instance, in the smart city deployment (Table VIII), under stringent 10 ms constraints, the algorithm decides on consistent 2 ms delay thresholds across all VNFs, achieving a cumulative E2E delay of exactly 10 ms with a minimum guarantee value of 0.58. As the E2E threshold relaxes to 45 ms, the algorithm exploits this flexibility by allocating larger delay thresholds to downstream VNFs, resulting in improved guarantee values while maintaining QoS requirements. The smart factory deployment decisions (Table IX) demonstrate notably higher promise guarantee values, particularly in the access and initial VNF domains, consistently achieving guarantee values of 1.0 for $D_{\text{th}} = 2$ ms. This superior performance can be attributed to more network nodes that can provide higher guarantee values to the smart factory traffic which stems from the smart factory traffic having a less bursty nature compared to the other IoT applications. Interestingly, even under relaxed E2E constraints ($Q^i \geq 25$ ms), the algorithm maintains tight delay thresholds (2 ms) for the first three components of the service chain as these nodes can provide high guarantee value. Also, since the parameter w_{QoS} is set to 1, the E2E algorithm does not consider reducing the amount of resources, as long as Q^i is achieved. For the eHealth traffic profile (Table X) with stringent E2E requirements ($Q^i = 10$ ms), the algorithm achieves moderate guarantee values across the chain. As the E2E threshold increases ($Q^i = 45$ ms), the algorithm demonstrates adaptive behavior by gradually increasing delay thresholds along the service chain (5, 5, 10, 10, and 10 ms), achieving higher performance guarantee values. Following the optimization of E2E promise chains, the selected promises associated with network nodes are implemented across the network infrastructure, where each designated node activates its corresponding D3QN policy associated with the chosen promises. To validate these deployment decisions, we conducted performance evaluations over a five-day testing period for each E2E delay threshold Q^i. The results of these deployments are captured through CDF analysis, presented in the rightmost column of Tables VIII–X. These CDF values, computed as $P(D_{\text{E2E}} \leq Q^i)$, where D_{E2E} represents the observed average E2E delay, provide a comprehensive measure of the deployment's ability to maintain the prescribed QoS constraints under real conditions. For smart city traffic, these values indicate the CDF values of D_{E2E} range from [0.96–1]. The smart factory profile demonstrates exceptional fulfillment of E2E delay thresholds with CDF values consistently above 0.99, while eHealth traffic shows a more gradual performance as the E2E threshold relaxes, ranging from 0.63 to 1.0.

2) *Comparative Study of E2E Delay Performance:* In this study, we compare the performance of the PPNO mechanism against three benchmark algorithms: a) the BDQN and b) two greedy algorithms. We employ a BDQN algorithm that is similar to the algorithm proposed in [45]. BDQN employs a sophisticated reinforcement learning architecture that simultaneously learns policies for multiple discrete actions, where the actions select the deployment decision of the access node and the placement of four VNFs. Moreover, the algorithm is trained for Q^i of 20 ms which is reflected in the reward similar to the formulation in (19). The state information includes the average service demand and the resource availability of each node in a previous one-hour time window. Moreover, we develop two greedy algorithms, GMIN and GMAX, where the first minimizes resource cost and the second maximizes service performance. Fig. 12 illustrates the CDF of E2E delay performance for the evaluated algorithms over a five-day validation period. The proposed PPNO mechanism demonstrated superior performance, achieving a high CDF value at the specified delay threshold of 20 milliseconds. In contrast, the BDQN algorithm exhibited suboptimal performance, with a CDF of approximately 0.6 at 20 ms and experiencing significant delay variations extending up to 180 ms. This performance degradation can be attributed to BDQN's limited temporal consideration window of one hour for resource demand and node availability, which proves insufficient for maintaining consistent delay performance below 20 ms across the five-day network slice lifecycle. The GMAX algorithm, which prioritizes nodes with maximum resource availability, achieved the lowest mean delay of 16 ms among all evaluated approaches. This performance can be attributed to its resource-abundant allocation strategy, ensuring sufficient capacity for varying service demands. Conversely, the GMIN algorithm, designed to minimize

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TABLE VIII
E2E NS DEPLOYMENT RESULT FOR THE SMART CITY TRAFFIC

E2E Threshold	Placement Decision: (D_{Th}, Guarantee)					E2E Promise (E2E delay, min guarantee)	CDF (E2E Threshold)
	Access	VNF1	VNF2	VNF3	VNF4		
10	(2, 0.97)	(2, 0.96)	(2, 0.95)	(2, 0.73)	(2, 0.58)	(10, 0.58)	0.96
15	(2, 0.97)	(2, 0.96)	(2, 0.95)	(5, 0.95)	(2, 0.58)	(13, 0.58)	0.99
20	(2, 0.97)	(2, 0.96)	(2, 0.95)	(5, 0.95)	(5, 0.79)	(16, 0.79)	1
25	(5, 0.89)	(5, 1.0)	(5, 0.99)	(5, 0.95)	(5, 0.79)	(25, 0.79)	0.97
35	(5, 0.89)	(5, 1.0)	(5, 0.99)	(10, 0.8)	(10, 0.8)	(30, 0.8)	1
45	(5, 0.89)	(5, 1.0)	(10, 1.0)	(10, 0.96)	(10, 0.8)	(40, 0.8)	1

TABLE IX
E2E NS DEPLOYMENT RESULT FOR THE SMART FACTORY TRAFFIC

E2E Threshold	Placement Decision: (D_{Th}, Guarantee)					E2E Promise (E2E delay, min guarantee)	CDF (E2E Threshold)
	Access	VNF1	VNF2	VNF3	VNF4		
10	(2, 1)	(2, 1)	(2, 1)	(2, 0.96)	(2, 0.64)	(10, 0.64)	0.99
15	(2, 1)	(2, 1)	(2, 1)	(5, 0.95)	(5, 0.76)	(13, 0.76)	1
20	(2, 1)	(2, 1)	(2, 1)	(10, 0.95)	(10, 0.8)	(18, 0.8)	0.99
25	(2, 1)	(2, 1)	(2, 1)	(10, 0.95)	(10, 0.8)	(18, 0.8)	1
35	(2, 1)	(2, 1)	(2, 1)	(10, 0.96)	(10, 0.8)	(26, 0.8)	1
45	(2, 1)	(2, 1)	(2, 1)	(10, 0.96)	(10, 0.8)	(26, 0.8)	1

TABLE X
E2E NS DEPLOYMENT RESULT FOR THE EHEALTH TRAFFIC

E2E Threshold	Placement Decision: (D_{Th}, Guarantee)					E2E Promise (E2E delay, min guarantee)	CDF (E2E Threshold)
	RAN	VNF1	VNF2	VNF3	VNF4		
10	(2, 0.97)	(2, 0.96)	(2, 0.95)	(2, 0.86)	(2, 0.58)	(10, 0.58)	0.63
15	(2, 0.99)	(2, 0.99)	(2, 0.98)	(2, 0.86)	(5, 0.76)	(13, 0.76)	0.91
20	(2, 0.99)	(2, 0.99)	(2, 0.98)	(5, 0.99)	(5, 0.76)	(16, 0.76)	0.97
25	(5, 0.99)	(5, 1.0)	(5, 0.99)	(5, 0.92)	(5, 0.76)	(25, 0.76)	0.95
35	(5, 0.99)	(5, 1.0)	(5, 0.99)	(10, 0.95)	(10, 0.77)	(30, 0.77)	0.99
45	(5, 0.99)	(5, 1.0)	(10, 1.0)	(10, 0.92)	(10, 0.77)	(40, 0.77)	1

resource consumption by selecting nodes with minimal available resources, resulted in substantial performance deterioration, with delays reaching up to 620 ms, highlighting the tradeoff between resource efficiency and service quality.

3) *Comparative Study of E2E Cost Performance:* The cost efficiency analysis, depicted in Fig. 13, reveals the resource optimization capabilities of each approach. As expected, the GMAX mechanism resulted in the highest resource cost and GMIN resulted in the lowest resource cost. The proposed PPNO principle demonstrated remarkable efficiency, achieving a 40% reduction in resource costs compared to GMAX while requiring only 10% additional resources relative to GMIN. Crucially, the PPNO maintained the stringent 20 ms delay threshold throughout the operation. While the BDQN algorithm achieved resource costs comparable to PPNO, its limited temporal analysis window of one hour led to suboptimal node selection in the service chain, resulting in excessive packet delays despite seemingly efficient resource utilization. The superior performance of the PPNO mechanism can be attributed to its implementation of adaptive resource allocation policies at each node, enabling dynamic resource optimization while maintaining delay requirements. This approach yielded the most favorable balance between delay threshold compliance and resource efficiency among all evaluated algorithms.

E. Soft Network Control Study

To evaluate the efficacy of the H-SliceMaster's SNC mechanism in implementing operator-based intents, we conducted a study analyzing system behavior across diverse IoT environments. The study focused on a representative operator intent: Minimize service deployment on computational nodes A, B, C, and D when feasible. This intent exemplifies common operational scenarios such as planned maintenance, software updates, or anticipated traffic surges in specific geographical regions that require traffic offloading and migration.

The IBF implements this intent through strategic price modulation of the specified nodes. We examined three distinct priority levels—low, medium, and high—corresponding to price increments of 1%, 10%, and 20%, respectively. These modified pricing parameters are subsequently communicated to the NSMF via NSSnaps.

The experimental setup encompassed four IoT environments (E1–E4), each characterized by a unique distribution of service categories, as detailed in Table XI. For instance, environment E1 predominantly consists of time-critical services (70%), with equal distributions (10%) of time-sensitive, delay-aware, and best-effort services. Time-critical services assigned a cost weight (w_{cost}) of 0, typically represent essential security or safety-critical applications where performance requirements supersede resource cost considerations.

Fig. 14 illustrates the resource utilization patterns across the specified nodes for varying environments and intent priorities. The baseline measurement, before intent implementation,

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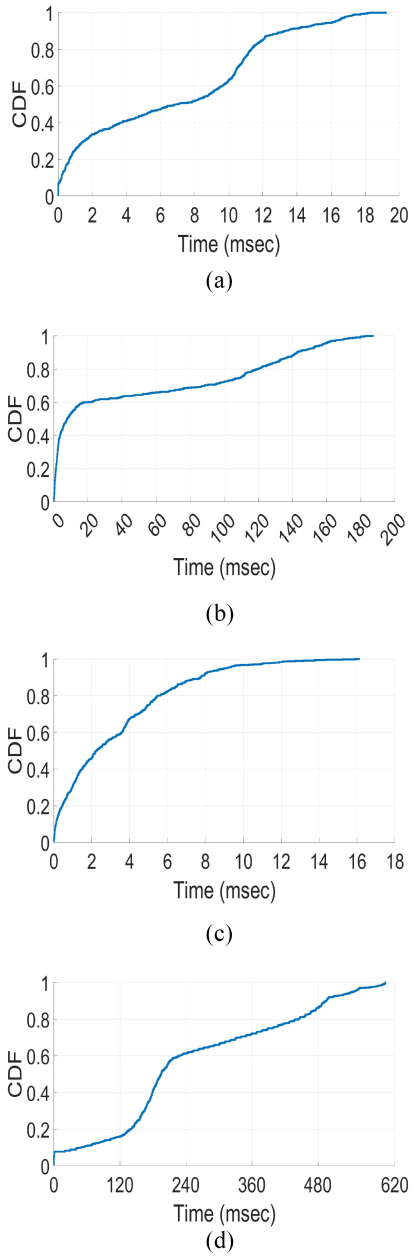


Fig. 12. Delay performance of (a) PPNO, (b) BDQN, (c) GMAX, and (d) GMIN.

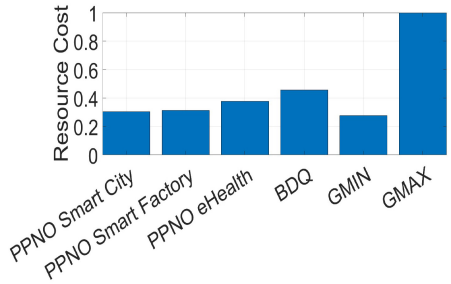


Fig. 13. Normalized resource cost of E2E algorithms.

showed 100% utilization across all nodes, e.g., all nodes are selected for VNF deployment. The results demonstrate that increasing intent priority consistently led to reduced resource utilization, validating the effectiveness of the SNC approach in enforcing operator intents. However, the magnitude of this effect varied significantly across environments. Environment E1, dominated by time-critical services, exhibited relatively high utilization (0.7–0.9) even under high-priority intents, while environments with a larger proportion of cost-sensitive services (higher w_{cost}) showed more dramatic reductions in utilization, reaching as low as 0.1.

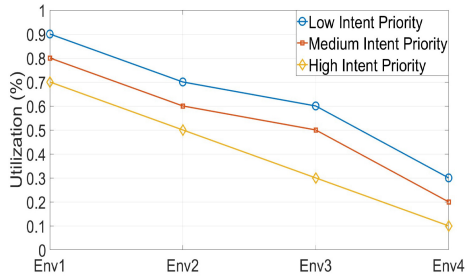


Fig. 14. System response to SNC mechanism.

TABLE XI
SERVICE DISTRIBUTION IN IOT ENVIRONMENTS

Service Type	w_{cost}	E1	E2	E3	E4
Time critical	0	0.7	0.5	0.3	0.1
Time-sensitive	0.5	0.1	0.1	0.2	0.1
Delay aware	0.7	0.1	0.1	0.1	0.1
Best effort	0.9	0.1	0.3	0.4	0.7

VII. Discussion on Limitations and Future Research Directions

A. Importance of Standardization of the PPNO Principle

The PoC simulation experiments of this work represent a critical breakthrough in demonstrating the transformative potential of the PPNO principle within E2E MO frameworks. By successfully satisfying stringent QoS requirements with robust privacy preservation, these experiments mark a significant milestone toward fully automated, privacy-preserving distributed network management systems.

The path forward necessitates a standardization strategy that addresses multiple critical dimensions. Foremost is the development of standardized PPLs that can be uniformly implemented across diverse system domains. Such standardization will enable E2E optimization, facilitating seamless interoperability for cross-domain coordination and integration.

Another important aspect is the standardization of identification and classification of TSPs. Establishing methodologies for TSP identification can unlock opportunities for collaborative learning, model updates, and intelligent policy development between distributed intelligent agents within system domains. This becomes particularly crucial in emerging IoT environments, where identification methods must encompass not just TSP characteristics, but also parameters such as network topology, scale, and contextual operational constraints that can

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affect the intelligent policy and the promises, e.g., monetary resource allocation costs.

B. Computational Complexity of the H-SliceMaster

Resource demands such as computational complexity of MFs within the H-SliceMaster framework is primarily determined by the algorithmic implementation strategies that influence resource requirements, with a clear distinction between heuristic rule-based approaches and advanced intelligent algorithms [52]. For instance, rule-based methods exhibit significantly lower computational and memory demands compared to an intelligent algorithm that necessitates continuous ML model training. However, in resource MO problems, ML-based approaches have demonstrated superior performance over rule-based methods [53]. While most research emphasizes the efficiency of optimization algorithms, further investigation is needed to assess the resource demands of these algorithms—whether ML-based or heuristic—when performing data analytics, IBN mechanisms such as intent resolution, and resource MO. Such studies can allow informed MO framework designs that comprehend the tradeoff, if any, between resource requirements and performance for different tasks within MFs.

Critically, the H-SliceMaster framework's design, does not mandate the deployment of specific algorithmic approaches, thereby offering adaptability across varied computational environments. This allows organizations to implement solutions ranging from lightweight rule-based systems to sophisticated intelligent algorithms, depending on specific operational requirements, available computational resources, and performance objectives.

C. Privacy-Preservation and Performance Tradeoffs

This study investigates the tradeoffs between privacy-preserving mechanisms of the H-SliceMaster and MO overhead. Through an implementation study, we demonstrate the practical feasibility of these frameworks across various network scales, highlighting key performance metrics that quantify the balance between privacy preservation and MO system efficiency, particularly in terms of bandwidth consumption and signaling latency.

Our findings indicate that the H-SliceMaster frameworks exhibit minimal bandwidth consumption, utilizing only 0.02% of total bandwidth availability in 5G networks when managing 200 network slices using the E2E H-SliceMaster. As the scale increases to 1000 network slices, the bandwidth overhead rises to approximately 0.4%. Notably, the implementation of the H-SliceMaster can adopt a hybrid approach, where certain network slices prioritize strong privacy preservation by leveraging all MFs of the E2E H-SliceMaster, while others employ the domain-based or NS-based frameworks with reduced privacy constraints. This hybrid design allows network operators to optimize framework configurations based on privacy requirements, resource consumption constraints, and availability considerations.

Furthermore, latency differentials highlight another critical tradeoff between privacy preservation and MO performance. The integration of privacy-preserving mechanisms within the H-SliceMaster frameworks necessitates additional MFs, which introduce signaling delays. Our analysis shows that these delays remain in the order of a few microseconds for MFs located within close geographical proximity. Moreover, practical implementations of the framework can facilitate direct communication channels between system domains and core network functions, ensuring ultralow-latency transmission for critical emergency or event-driven communications.

D. H-SliceMaster and Emerging Technologies

The H-SliceMaster framework represents a pivotal MO framework with significant potential for integration with cutting-edge security technologies. Future research directions can explore the integration of blockchain and FL as they emerge as particularly promising technological paradigms that could substantively enhance the framework's security, privacy, and collaborative capabilities.

Blockchain technology offers a transformative approach to authentication and access control within IoT networks, as demonstrated by recent research [54], [55]. Within the H-SliceMaster framework, a blockchain network could serve as a robust mechanism for broadcasting PPLs across participating system domains. The immutable nature of blockchain ledgers provides transparency and accountability, ensuring that system domains remain committed to their shared PPL agreements through publicly verifiable transactions.

The integration of FL within the IBFs presents another frontier of technological innovation. This approach enables collaborative ML model updates while maintaining stringent privacy preservation protocols. By distributing the learning process across multiple domains without centralizing raw data, FL addresses critical privacy concerns inherent in traditional ML methodologies.

VIII. Conclusion

This article has introduced the H-SliceMaster, a privacy-preserving I-IBNS framework for IoT systems. The framework addresses critical challenges in contemporary MO systems, particularly the complex balance between E2E optimization and privacy preservation in multiadministrator environments. Through the proposed PPNO principle, the H-SliceMaster enables system domains to maintain operational autonomy while participating in E2E resource optimization.

The framework's efficacy has been demonstrated through extensive simulations across diverse IoT environments, with results showing significant improvements in both performance and resource utilization. Specifically, the PPNO mechanism achieved around 30 to 50% reduction in resource costs compared to BDQN and greedy allocation strategies while maintaining stringent delay requirements. The framework's ability to support multiple TSPs was validated across smart city, smart factory, and eHealth applications, with E2E delay compliance rates exceeding 96% for most scenarios.

Furthermore, the proposed SNC mechanism has proven effective in implementing operator intents without compromising domain privacy, demonstrating varying degrees of

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responsiveness across different service distributions. This approach represents a significant advancement in IBN, offering a practical solution to balancing centralized control with domain autonomy.

Future research directions could include extending the framework to support different network slice requirements, investigating the impact of varying promise guarantee formulations on system performance, and developing more sophisticated price modulation strategies for the SNC mechanism.

REFERENCES

[1] K. Fizza, A. Banerjee, P. P. Jayaraman, N. Auluck, R. Ranjan, K. Mitra, and D. Georgakopoulos, "A survey on evaluating the quality of autonomic Internet of Things applications," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 1, pp. 567–590, 1st Quart., 2023.

[2] A. Pagès, F. Agraz, J. B. Caro, and S. Spadaro, "Machine learning-based multi-domain actuation orchestration in support of end-to-end service quality-assurance," *IEEE Trans. Netw. Service Manag.*, vol. 20, no. 3, pp. 2575–2586, Sep. 2023.

[3] S. Bhattacharjee, K. Alexandris, and T. Bauschert, "Multi-domain TSN orchestration & management for large-scale industrial networks," *IEEE Trans. Netw. Service Manag.*, vol. 21, no. 6, pp. 6480–6492, Dec. 2024.

[4] "Technical specification group services and system aspects; management and orchestration; network slicing; concepts, use cases and requirements; (Release 15), Version 15.0.0," 3GPP, Sophia Antipolis, France, Rep. TS 28.530, 2020, Accessed: Sep. 2024.

[5] P. Wyszkowski, J. Kienig, K. Zieliński, Ł. Czekierda, and M. Zawadzki, "Comprehensive tutorial on the organization of a standards-aligned network slice/subnet design process and opportunities for its automation," *IEEE Commun. Surveys Tuts.*, vol. 26, no. 2, pp. 1386–1445, 2nd Quart., 2024.

[6] A. Clemm, L. Ciavaglia, L. Z. Granville, and J. Tantsura, "Intent-based networking-concepts and definitions," Internet Eng. Task Force, RFC 9315, Oct. 2022.

[7] A. Leivadeas and M. Falkner, "A survey on intent-based networking," *IEEE IEEE Commun. Surveys Tuts.*, vol. 25, no. 1, pp. 625–655, 1st Quart., 2023.

[8] M. Xie, P. H. Gomes, J. Niemöller, and J. P. Waldemar, "Intent-driven management for multi-vertical end-to-end network slicing services," in *Proc. IEEE Globecom Workshops (GC Wkshps)*, 2022, pp. 1285–1291.

[9] R. Caldelli, P. Castoldi, M. Gharbaoui, B. Martini, M. Matarazzo, and F. Sciarrone, "On helping users in writing network slice intents through NLP and user profiling," in *Proc. IEEE 9th Int. Conf. Netw. Softw. (NetSoft)*, 2023, pp. 545–550.

[10] P. Doanis, T. Giannakas, and T. Spyropoulos, "Scalable end-to-end slice embedding and reconfiguration based on independent DQN agents," in *Proc. GLOBECOM IEEE Global Commun. Conf.*, 2022, pp. 3429–3434.

[11] A. Gharehgoli, A. Nouruzi, N. Mokari, P. Azmi, M. R. Javan, and E. A. Jorswieck, "AI-based resource allocation in end-to-end network slicing under demand and CSI uncertainties," *IEEE Trans. Netw. Service Manag.*, vol. 20, no. 3, pp. 3630–3651, Sep. 2023.

[12] J. Huang, C. Yang, S. Kou, and Y. Song, "A brief survey and implementation on ai for intent-driven network," in *Proc. 27th Asia Pac. Conf. Commun. (APCC)*, 2022, pp. 413–418.

[13] Z. Bashir, S. Kumar, V. Murgai, and S. Singh, "Cloud ran based privacy preserving federated cross domain anomaly detection in IoT devices logs," in *Proc. IEEE Globecom Workshops (GC Wkshps)*, 2023, pp. 1820–1825.

[14] H. Zhang, H. Zhou, and M. Erol-Kantarci, "Federated deep reinforcement learning for resource allocation in O-RAN slicing," in *Proc. GLOBECOM IEEE Global Commun. Conf.*, 2022, pp. 958–963.

[15] N. Slamnik-Kriještorac, G. M. Yilma, M. Liebsch, F. Z. Yousaf, and J. M. Marquez-Barja, "Collaborative orchestration of multi-domain edges from a connected, cooperative and automated mobility (CCAM) perspective," *IEEE Trans. Mobile Comput.*, vol. 22, no. 4, pp. 2001–2020, Apr. 2023.

[16] J. S. Choi, S. J. Chun, and S. Lee, "Hierarchical distributed overarching architecture of decoupled federation and orchestration frameworks for multidomain NFV MANOs," *IEEE Commun. Mag.*, vol. 60, no. 9, pp. 68–74, Sep. 2022.

[17] F. Z. Yousaf, V. Sciancalepore, M. Liebsch, and X. Costa-Perez, "MANOaaS: A multi-tenant NFV MANO for 5G network slices," *IEEE Commun. Mag.*, vol. 57, no. 5, pp. 103–109, May 2019.

[18] H. Bai, Y. Zhang, Z. Zhang, and S. Yuan, "Latency equalization policy of end-to-end network slicing based on reinforcement learning," *IEEE Trans. Netw. Service Manag.*, vol. 20, no. 1, pp. 88–103, Mar. 2023.

[19] M. A. Habibi et al., "Toward an open, intelligent, and end-to-end architectural framework for network slicing in 6G communication systems," *IEEE Open J. Commun. Soc.*, vol. 4, pp. 1615–1658, 2023.

[20] E. Chirivella-Perez, P. Salva-Garcia, I. Sanchez-Navarro, J. M. Alcaraz-Calero, and Q. Wang, "E2E network slice management framework for 5G multi-tenant networks," *J. Commun. Netw.*, vol. 25, no. 3, pp. 392–404, 2023.

[21] B. Yu, X. Chi, S. Li, X. Liu, and S. Ren, "Martingale-based URLLC slice customization for the provisioning of reliability with regard to end-to-end latency," *IEEE Internet Things J.*, vol. 11, no. 1, pp. 1311–1327, Jan. 2024.

[22] J. Liu, B. Zhao, M. Shao, Q. Yang, and G. Simon, "Provisioning optimization for determining and embedding 5g end-to-end information centric network slice," *IEEE Trans. Netw. Service Manag.*, vol. 18, no. 1, pp. 273–285, Mar. 2021.

[23] Q. Liu, N. Choi, and T. Han, "Deep reinforcement learning for end-to-end network slicing: Challenges and solutions," *IEEE Netw.*, vol. 37, no. 2, pp. 222–228, Mar./Apr. 2023.

[24] M. Chahbar, G. Diaz, A. Dandoush, C. Cérin, and K. Ghoumid, "A comprehensive survey on the E2E 5G network slicing model," *IEEE Trans. Netw. Service Manag.*, vol. 18, no. 1, pp. 49–62, Mar. 2021.

[25] H. Li et al., "Slice-based service function chain embedding for end-to-end network slice deployment," *IEEE Trans. Netw. Service Manag.*, vol. 20, no. 3, pp. 3652–3672, Sep. 2023.

[26] M. Masoudi, z. T. Demir, J. Zander, and C. Cavdar, "Energy-optimal end-to-end network slicing in cloud-based architecture," *IEEE Open J. Commun. Soc.*, vol. 3, pp. 574–592, 2022.

[27] H. Cao et al., "Toward tailored resource allocation of slices in 6G networks with softwarization and virtualization," *IEEE Internet Things J.*, vol. 9, no. 9, pp. 6623–6637, May 2022.

[28] W.-K. Chen, Y.-F. Liu, A. De Domenico, Z.-Q. Luo, and Y.-H. Dai, "Optimal network slicing for service-oriented networks with flexible routing and guaranteed E2E latency," *IEEE Trans. Netw. Service Manag.*, vol. 18, no. 4, pp. 4337–4352, Dec. 2021.

[29] X. Li et al., "5Growth: An end-to-end service platform for automated deployment and management of vertical services over 5G networks," *IEEE Commun. Mag.*, vol. 59, no. 3, pp. 84–90, Mar. 2021.

[30] M. Xie et al., "Practically deploying multiple vertical services into 5G networks with network slicing," *IEEE Netw.*, vol. 36, no. 1, pp. 32–39, Jan./Feb. 2022.

[31] A. A. Abdellatif, A. Abo-Eleneen, A. Mohamed, A. Erbad, N. V. Navkar, and M. Guizani, "Intelligent-slicing: An AI-assisted network slicing framework for 5G-and-beyond networks," *IEEE Trans. Netw. Service Manag.*, vol. 20, no. 2, pp. 1024–1039, Jun. 2023.

[32] Y. Yang et al., "6G network AI architecture for everyone-centric customized services," *IEEE Netw.*, vol. 37, no. 5, pp. 71–80, Sep. 2023.

[33] W. Wu et al., "AI-native network slicing for 6G networks," *IEEE Wireless Communications*, vol. 29, no. 1, pp. 96–103, Feb. 2022.

[34] H. Chergui, A. Ksentini, L. Blanco, and C. Verikoukis, "Toward zero-touch management and orchestration of massive deployment of network slices in 6G," *IEEE Wireless Commun.*, vol. 29, no. 1, pp. 86–93, Feb. 2022.

[35] Y. Wu, H.-N. Dai, H. Wang, Z. Xiong, and S. Guo, "A survey of intelligent network slicing management for industrial IoT: Integrated approaches for smart transportation, smart energy, and smart factory," *IEEE IEEE Commun. Surveys Tuts.*, vol. 24, no. 2, pp. 1175–1211, 2nd Quart., 2022.

[36] L. Velasco et al., "End-to-end intent-based networking," *IEEE Commun. Mag.*, vol. 59, no. 10, pp. 106–112, Oct. 2021.

[37] C. Yang, X. Mi, Y. Ouyang, R. Dong, J. Guo, and M. Guizani, "Smart intent-driven network management," *IEEE Commun. Mag.*, vol. 61, no. 1, pp. 106–112, Jan. 2023.

[38] R. Dong, C. Yang, J. Guo, J. Zhang, and J. Huang, "An intent-driven next-generation mobile communication management system," in *Proc. IEEE 22nd Int. Conf. Commun. Technol. (ICCT)*, 2022, pp. 1270–1275.

[39] M. Sebrechts, B. Volckaert, F. De Turck, K. Yang, and M. Al-Naday, "Fog native architecture: Intent-based workflows to take cloud native toward the edge," *IEEE Commun. Mag.*, vol. 60, no. 8, pp. 44–50, Aug. 2022.

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[40] J. Zhang, C. Yang, A. Anpalagan, R. Dong, and L. Zhang, "IDSM: Intent-driven slice management and maintenance for 6G RANs," in *Proc. IEEE 22nd Int. Conf. Commun. Technol. (ICCT)*, 2022, pp. 1244–1249.
[41] J. C. Cisneros, S. E. P. Hernández, J. C. P. Sansalvador, L. M. R. Henríquez, S. Yangui, and K. Drira, "Coordination-free multi-domain NFV orchestration for consistent VNF forwarding graph reconfiguration," *IEEE Trans. Netw. Service Manag.*, vol. 19, no. 4, pp. 5133–5151, Dec. 2022.
[42] A. Dalgkitsis et al., "SCHE2MA: Scalable, energy-aware, multidomain orchestration for beyond-5G URLLC services," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 7, pp. 7653–7663, Jul. 2023.
[43] G. M. Almeida et al., "RIC-O: Efficient placement of a disaggregated and distributed RAN intelligent controller with dynamic clustering of radio nodes," *IEEE J. Sel. Areas Commun.*, vol. 42, no. 2, pp. 446–459, Feb. 2024.
[44] M. Iwamoto, A. Suzuki, and M. Kobayashi, "Optimal VNF scheduling for minimizing duration of QoS degradation," in *Proc. IEEE 20th Consum. Commun. Netw. Conf. (CCNC)*, 2023, pp. 855–858.
[45] S. Malektaji, M. Rayani, A. Ebrahimzadeh, V. M. Raee, H. Elbiaze, and R. H. Glitho, "Dynamic joint VNF forwarding graph composition and embedding: A deep reinforcement learning framework," *IEEE Trans. Netw. Service Manag.*, vol. 20, no. 4, pp. 4615–4633, Dec. 2023.
[46] D. H. Hussein and M. Ibnkahla, "A novel mathematical framework for modeling application-specific IoT traffic," *IEEE Internet Things J.*, vol. 11, no. 2, pp. 2364–2381, Jan. 2024.
[47] D. Bega, M. Gramaglia, M. Fiore, A. Banchs, and X. Costa-Perez, "Aztec: Anticipatory capacity allocation for zero-touch network slicing," in *Proc. IEEE INFOCOM Conf. Comput. Commun.*, 2020, pp. 794–803.
[48] "Study on RAN improvements for machine-type communications; (Release 10)," 3GPP, Sophia Antipolis, France, Rep. 37.868, 2012.
[49] H. Zhang et al., "Large-scale measurements and optimizations on latency in edge clouds," *IEEE Trans. Cloud Comput.*, vol. 12, no. 4, pp. 1218–1231, Oct.–Dec. 2024.
[50] D. H. Hussein and M. Ibnkahla, "An IoT traffic modeling framework and its application to autonomous edge scaling," in *Proc. GLOBECOM IEEE Global Commun. Conf.*, 2022, pp. 5656–5661.
[51] M. Piro and R. Wessäly. "Sndlib 1.0 survivable network design library." Accessed: Apr. 10, 2024. https://sndlib.put.poznan.pl/home.action
[52] G. Drainakis, P. Pantazopoulos, K. V. Katsaros, V. Sourlas, and A. Amditis, "On the resource consumption of distributed ML," in *Proc. IEEE Int. Symp. Local Metropol. Area Netw. (LANMAN)*, 2021, pp. 1–6.
[53] D. H. Hussein and M. Ibnkahla, "Intelligent and autonomous edge slicing for IoT systems," *IEEE Internet Things J.*, early access, Feb. 26, 2025, doi: 10.1109/JIOT.2025.3546124.
[54] D. H. Hussein, R. Anbarasu, A. Matrawy, and M. Ibnkahla, "Towards a decentralized access control system for IoT platforms based on blockchain technology," in *Proc. Int. Symp. Netw., Comput. Commun. (ISNCC)*, 2020, pp. 1–6.
[55] D. H. Hussein, E. Houlahan, A. Janelle-Goode, T. Lumsden, and M. Ibnkahla, "A blockchain-based dual identity management and authentication framework for IoT networks," in *Proc. IEEE 10th World Forum Internet Things (WF-IoT)*, 2024, pp. 544–549.



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